An interface framework and adequacy standard when generative turbulence models hit the wall

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Abstract

Generative models are gaining attention in fluid mechanics amid hopes of a step change in turbulence modelling from whole-field reconstruction. Yet they inherit the century-old difficulty of computational fluid dynamics: representing the near-wall region. It occupies little volume yet sets drag, separation, wall-generated vorticity and loads. Volume-averaged training and validation objectives weight its cells by count, not dynamical leverage. We identify why the commonly used wall-closure interface fails and build one that works. Expanding predicted wall stress into an equilibrium-law velocity band destroys the information; through a conditioning slot the generator was trained to read, the stress propagates. Across a bluff-body cube array, attached channel and separating periodic hill—the cube and hill deliberately difficult engineering flows—cube viscous wall-force error falls by two-fifths, the channel recovers three-quarters of the exact-traction benefit, and the hill transmits in one of two generative families. The result is an interface framework and adequacy standard: an equal-footing slot that delivers any wall model's stress to the generator undistorted, and an oracle control certifying whether a generative family detects wall information, offered to refocus generative turbulence modelling on the wall. Evaluations are offline and a priori. Code and controls are available at https://github.com/JianouJiang/generative-wall-turbulence.

1 Introduction

Modern turbulence modelling has changed its mathematics—from Prandtl's boundary-layer construction, through Reynolds-averaged and large-eddy simulation, to learned generative fields—but it keeps reaching the same physical obstacle: the wall [1–4]. A geometrically thin near-wall layer sets skin friction; reversal of wall shear accompanies separation; and the resulting vorticity, shear layers and wakes organise flow well beyond it [2, 5]. Coherent near-wall motions sustain turbulence, while outer scales partly predict them at high Reynolds number [6–11]. Resolving this small region dominates simulation cost at engineering Reynolds numbers [12, 13]; conventional simulation has therefore spent decades coupling unresolved wall physics to a resolved outer flow [14–16]. Generative models do not escape that physics simply by learning the field.

Machine learning now supplies data-driven closures, solver acceleration and learned operators [17–23]. Generative families represent whole-field distributions for inflow, wakes, reconstruction, super-resolution and subfilter closure [24–36], with conditional variants reconstructing fields from partial observations [37–41]. Yet two properties create a wall-specific failure mode: neural networks preferentially fit low-frequency, large-scale content [42, 43], and a volume-averaged loss weights a thin wall band by cell count rather than by dynamical leverage. A boundary-localised error can therefore contribute little to a global objective while materially changing drag or separation (Supplementary Table 13). Validation inherits the same weighting: a volume-averaged score can certify a reconstruction whose implied wall stress is wrong. Wall treatment is not a niche refinement of a whole-field model; it limits what that model can establish wherever surface forces matter.

Representative whole-field formulations are not universally wall-blind: Gao *et al.* generate wall-bounded fields [44]; CoNFiLD includes periodic-hill, rough-wall and channel cases and documents larger discrepancies near the wall [45]; Oommen *et al.* address super-resolution, forecasting and sparse reconstruction [43, 46–48]. Our concern is not that wall-bounded examples are absent, and we are not claiming that machine learning universally ignores walls. Yet whole-field objectives do not by themselves give a thin wall region its engineering weight, and none has an explicit wall-closure-to-generator interface.

Wall-focused work provides important counterexamples: early neural models of near-wall turbulence [49], turbulent fields estimated from wall shear and pressure [50–52], low-order wall-bounded surrogates [53, 54], learned synthetic inflow [29, 55, 56], conditional flow matching from wall observations [57] and latent-diffusion wall-pressure reconstruction [58]. In parallel, wall models represent the unresolved near-wall layer for large-eddy simulation [59–63], including machine-learning wall models [64, 65] extended to pressure-gradient and separated regimes [66, 67]. The strands answer different questions—which field is compatible with supplied wall data, versus which unresolved surface quantity can be supplied to a resolved computation—and neither role substitutes for the other in a coupled simulation.

Two things are missing between them. There is no tested *interface*: no whole-field generative formulation takes its wall boundary from a closure—generative models have produced closure terms [36], not consumed a wall model's output—and no wall closure has been judged by what a generator can propagate from it. And there is no *standard*

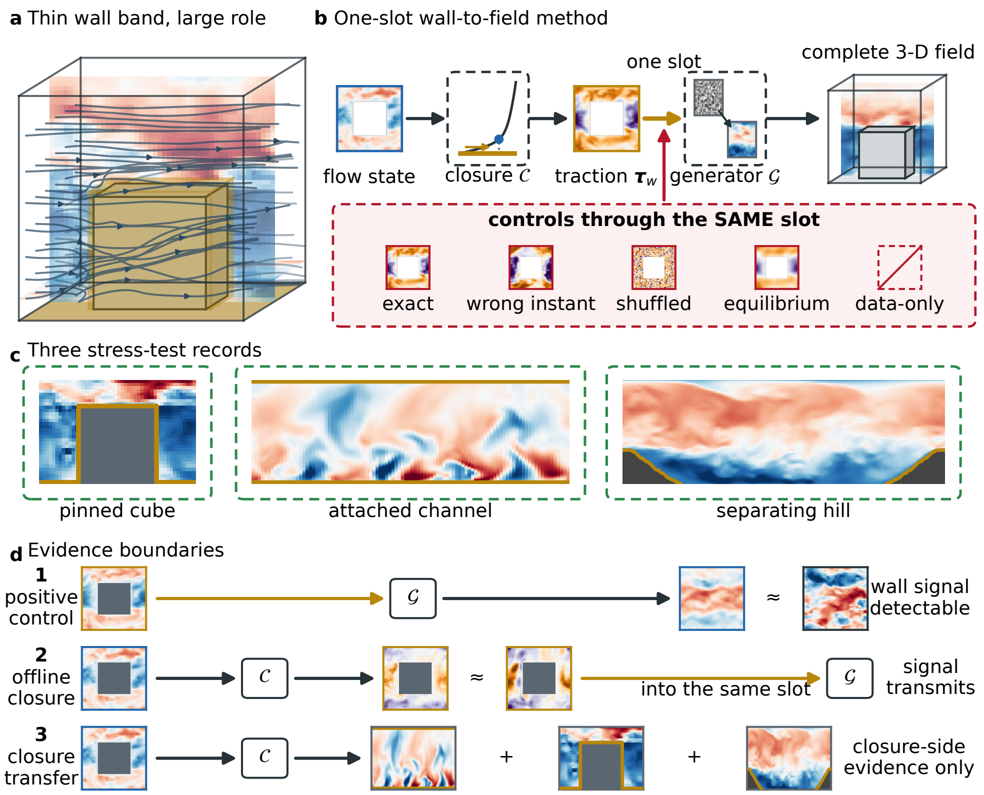

Fig. 1 Why generative turbulence models need an explicit wall interface. a, The thin supplied near-wall band (gold) wraps the floor and cube while its influence is assessed through the surrounding three-dimensional field. **b**, The one-slot method: matching-height flow state enters closure $\mathcal{C}$, whose signed traction $\boldsymbol{\tau}_w$ conditions generator $\mathcal{G}$; every wall model and control enters through this same slot. **c**, Three deliberately distinct stress tests and their qualitative outcomes: the pinned cube shows field and wall-load response, the attached channel reaches positive absolute skill, and the separating hill transmits only in diffusion and remains indicative. **d**, Evidence boundaries: the oracle band is the positive control, the fitted closure tests offline composition (Figs 2–8), and separate frozen models provide closure-side transfer only. No row represents solver-coupled deployment.

for testing one—no agreed way to show that a record can detect wall information at all before it is used to argue that wall information does or does not matter. Without that, a generative model responding to nothing and a wall channel carrying nothing produce the same measurement.

This paper supplies both through three linked tests of one integrated closure-to-generative-field system: diagnose failure of an equilibrium-band adapter, repair the interface by conditioning directly on signed surface traction, and standardise the positive control needed to establish that a record and generator can detect wall information. Every wall model enters one conditioning slot without generator-side exposure, so models are scored on equal footing. We also measure the fidelity needed for reconstruction benefit and provide a leakage-free public-corpus protocol. Figure 1 contrasts thin support with field influence (**a**), shows the one-slot method (**b**), identifies the stress tests and outcomes (**c**), and separates oracle detectability, offline composition and closure-only transfer (**d**). All evaluations are offline and a priori: no momentum equation is advanced, and oracle observations, closure outputs and solver deployment are distinct evidence.

The records are deliberately distinct: a geometrically pinned aligned-cube array, an attached channel and pressure-gradient-induced separation over a periodic hill. The cube and hill are difficult engineering regimes, not flattering selections. Expanding predicted stress into an equilibrium-law velocity band destroys the signal, whereas direct signed traction transmits 0.63 of a matched oracle ceiling. Closure-predicted traction then improves the source-excluded cube field by $+0.061\,[0.057, 0.064]$ and cuts inferred viscous wall-force error by two-fifths. The channel reaches positive absolute skill through the equal-footing slot (Fig. 8). On the hill, diffusion gains $+0.062\,[+0.017, +0.106]$ but carries roughly three effective samples; the preregistered flow-matching arm fails its positive control. Complete-volume cube skill remains negative.

Records enter the claim set only after source validation: one public record whose offline composition initially appeared favourable is excluded because its primary source identifies it as a two-dimensional irregular pipe with stochastic forcing [45], contradicting the wall geometry assumed (Supplementary Note 8). Companion results evaluate the closure alone and are preserved byte-for-byte without revalidation.

2 Results

2.1 An oracle near-wall band changes the field outside it

The foreground flow is an aligned-cube large-eddy simulation (LES) [68–70] with plan solidity $\lambda_p = 0.25$ and $Re_h = 5000$—exposed floor, ceiling and cube surfaces rather than a stack of homogeneous planes (native first-cell $y^+ \leq 1.92$; Methods). Of 1,101 post-spin-up fields, 660 train, 98 are excluded, and 160 temporally correlated, thinned fields are evaluated from the chronologically separated remainder. "LES reference" denotes this numerical record, not ground truth: the case has no experimental benchmark or grid/filter study.

The supplied band contains 14,008 of 207,360 fluid cells (6.8%; 4.5% excluding two computational-lid strips), so the complete endpoint scores 93.24% of the fluid domain; near and farther endpoints split the unsupplied cells at $d/h = 0.5$, a geometric label, not an equilibrium-layer claim. The intervention is rich—a dense oracle velocity band, not wall pressure, shear or a closure output—and tests whether a conditional generator changes outside the supplied support given time-aligned wall information. The far-time control shifts that band by ≈ 1.40 integral times at unchanged support and density.

The chronological allocation was recorded before any wall-arm output existed (Supplementary Note 1). Three sequential configurations of the *same* three-dimensional U-Net were run, called M0, M1 and M2. They differ only in size and budget: M0 width 24 and 30,000 updates; M1 width 48, 90,000 updates and a longer sampler; M2 differs

Table 1 Valid retained controls that delimit attribution. All values are fluctuation R^2 in the complete/near/farther-from-wall regions. These later controls use the same chronological allocation but different estimators or training protocols from M0; they are scope tests, not a clean numerical leaderboard.

Estimator and information	Complete	Near	Farther
Linear/Wiener, no band	−0.00008	−0.00001	−0.00018
Linear/Wiener, near-wall band	−0.05068	+0.02997	−0.13327
Deterministic direct band, seed 2234	+0.05523	+0.15647	−0.04842
Deterministic direct band, seed 3234	+0.05803	+0.16121	−0.04762
Nearest training band, associated field	−0.80255	−0.73251	−0.87430
M0 learned absent-band arm	−0.07240	−0.07669	−0.06804

from M1 only in using circular padding in the two periodic directions. What separates them evidentially matters more: M0 provided the only pre-output contact with the evaluation interval, so it is the first-contact comparison, while M1 and M2 reused that interval and are development sensitivities, not independent confirmations.

On the complete support-excluded volume M0 gives aligned-band $R^2_{\mathrm{fluct}} = 0.103$, against -0.072 for the absent-band branch of the same network and -0.204 for a far-time band: paired gains of $+0.175$ and $+0.307$, larger near the wall, with farther-from-wall absolute skill negative (Supplementary Table 2). M1 and M2 preserve the ordering (complete gains 0.174, 0.197), but the farther-from-wall paired interval crosses zero for M1, so no model-independent outer effect is established.

Table 1 delimits attribution: no estimator matched to M0's architecture, budget and selection protocol was run, so generator superiority is not established; a frozen-record retrieval diagnostic is strongly negative in every region, rejecting literal retrieval though not more flexible memorisation; and a support-contaminated interior comparison is quarantined (Supplementary Note 1). The selected M2 endpoint is not an untouched held-out test.

Figure 2 makes the intervention spatially concrete. Panel **a** fixes the first evaluated M2 target before output inspection and pairs its contemporaneous band with an equal-support donor 172 stored fields later. Panels **b**,**c** compare the LES section, three eight-sample arm means and their absolute errors on common scales. The aligned arm reduces error most visibly beside the floor and cube. The mean profiles remain similar (**d**), but the wall-normal profile resolves the local difference (**e**): aligned has the lowest near-wall error and complete-volume RMSE, 0.112 against 0.128 absent and 0.136 far-time. This is a reconstruction benefit rather than merely a shifted mean profile. Because it shows one target from development-contacted M2, the display illustrates rather than independently tests the aggregate result.

Figure 3 separates mechanism from evidential claim. Panels **a**,**b** map where correct-time information reduces squared error: improvement clusters around the floor and cube but is not uniform in the outer flow. Panel **c** holds the M2 U-Net and eight noise seeds fixed while only wall information changes; panel **d** removes supplied support before defining complete, near and farther scores, and distinguishes absolute fluctuation R^2 from paired ΔR^2. Panels **e**,**f** then return to first-contact M0: aligned skill

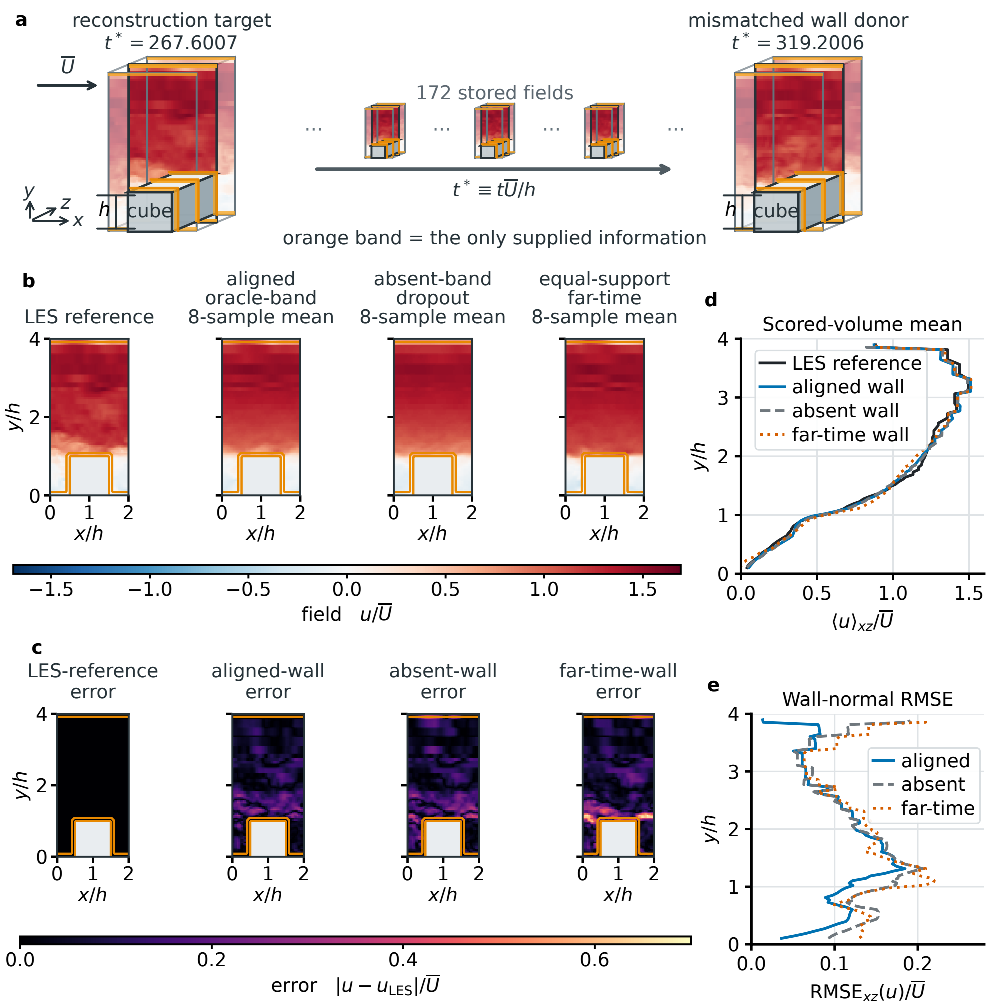

Fig. 2 Representative aligned-cube reconstruction. **a**, First evaluated M2 LES target (record 758, $t\overline{U}/h = 267.6$), fixed in advance by the protocol rather than chosen visually, with three retained intermediate fields and far-time wall donor (record 930); ellipses denote omitted fields. Each cutaway shows three x–y sections (dark: mid-span); orange marks the supplied near-wall band. **b**, Mid-span LES reference and aligned-band, absent-band and far-time eight-sample means. **c**, Absolute errors. **d**, Scored-volume mean profile. **e**, Wall-normal root-mean-square error (RMSE) profile; complete-volume RMSE 0.112 (aligned), 0.128 (absent), 0.136 (far-time). One target is shown; aggregate results use 160 temporally correlated three-component fields.

is 0.103 complete, 0.243 near and −0.040 farther, while gains over absent/far-time are 0.175/0.307, 0.320/0.564 and 0.028/0.044, respectively. Thus the wall signal is strongest near its support; the small positive farther contrasts do not produce positive outer absolute skill. Only terminal M0 aggregates survive, so points are shown without unreplayable intervals.

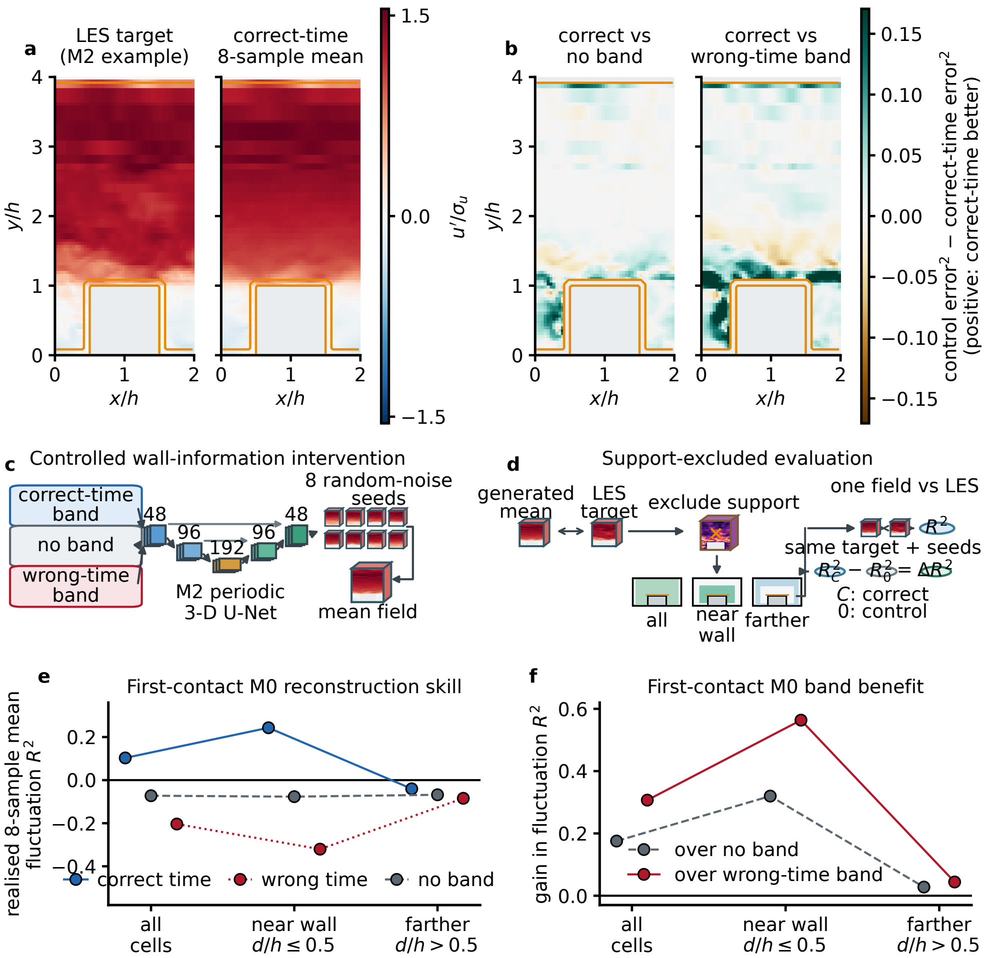

Fig. 3 Correct-time near-wall information acts chiefly near its supplied support. **a**, M2 mid-span LES target and eight-sample mean; orange marks supplied cells. **b**, Local error reduction against no-band and wrong-time controls; green favours correct time. **c**, Three matched arms traverse the frozen 3-D U-Net; larger cube, their eight-seed mean. **d**, All-cell, near-wall and farther masks; R^2 scores one arm, ΔR^2 a matched pair. Aligned, absent and far-time here score 0.120, −0.077, −0.314; near-wall aligned skill 0.284. **e**, M0 field-to-LES R^2. **f**, Correct-time gain over controls. M0 points are terminal aggregates without replayable intervals, but anchor interpretation as the only pre-contact comparison.

Post-selection distributional endpoint. A registered farther-region energy-score endpoint, frozen after M2 selection, favours the aligned arm over both controls (0.522 against 0.527 and 0.548)—a conditional post-selection sensitivity. **Its physical corroboration is null**: both valid statistic families fail to beat both controls, and the registered coverage family was withdrawn as invalid at eight members (Supplementary Note 4).

2.2 An equilibrium velocity reconstruction destroys the interface

A wall closure cannot supply a dense oracle velocity band; it supplies a signed two-component wall traction. We first executed the standard adapter—expand that traction into a velocity band through an equilibrium law and impose it. All arms use model M1, byte-identical to Section 2.1 in generator, targets, sampler noise and scoring masks; only the band payload changes.

Two raster defects change how the published band should be read: supplied cells carrying no information, and duplicates of scored cells. Restricting the band to the physical walls removes both; the leak-free arm still improves on the absent branch by $+0.102\,[0.093, 0.109]$, so the computational ceiling accounts for about 42% of the published $+0.174$ gain (Supplementary Note 3).

Against that matched ceiling the traction interface fails (Fig. 4). The target-velocity equilibrium inversion through the frozen lift scores -0.117 on the complete volume, a change of $-0.032\,[-0.048, -0.021]$ relative to supplying nothing, and its transmission ratio against the matched ceiling is $-0.320\,[-0.517, -0.194]$. The controls locate the failure: the traction is not information-free—the instantaneous inversion beats the mean wall load and a far-time traction ($+0.199\,[0.162, 0.239]$)—yet every lifted arm sits below the absent branch (Fig. 4b,c). The a-priori diagnostic explains why: the lifted band reproduces the true band with fluctuation $R^2 = -0.109 \pm 0.185$, worse than the band's own mean, and the sampler clamps that error into the field along the whole trajectory (Eq. (6)).

This bounded negative falsifies only the composition tested. Two confounds live in the adapter, not in the wall information: the lift is an equilibrium model of an instantaneous field, and every lifted arm is outside the frozen generator's conditioning distribution. The next subsections remove it.

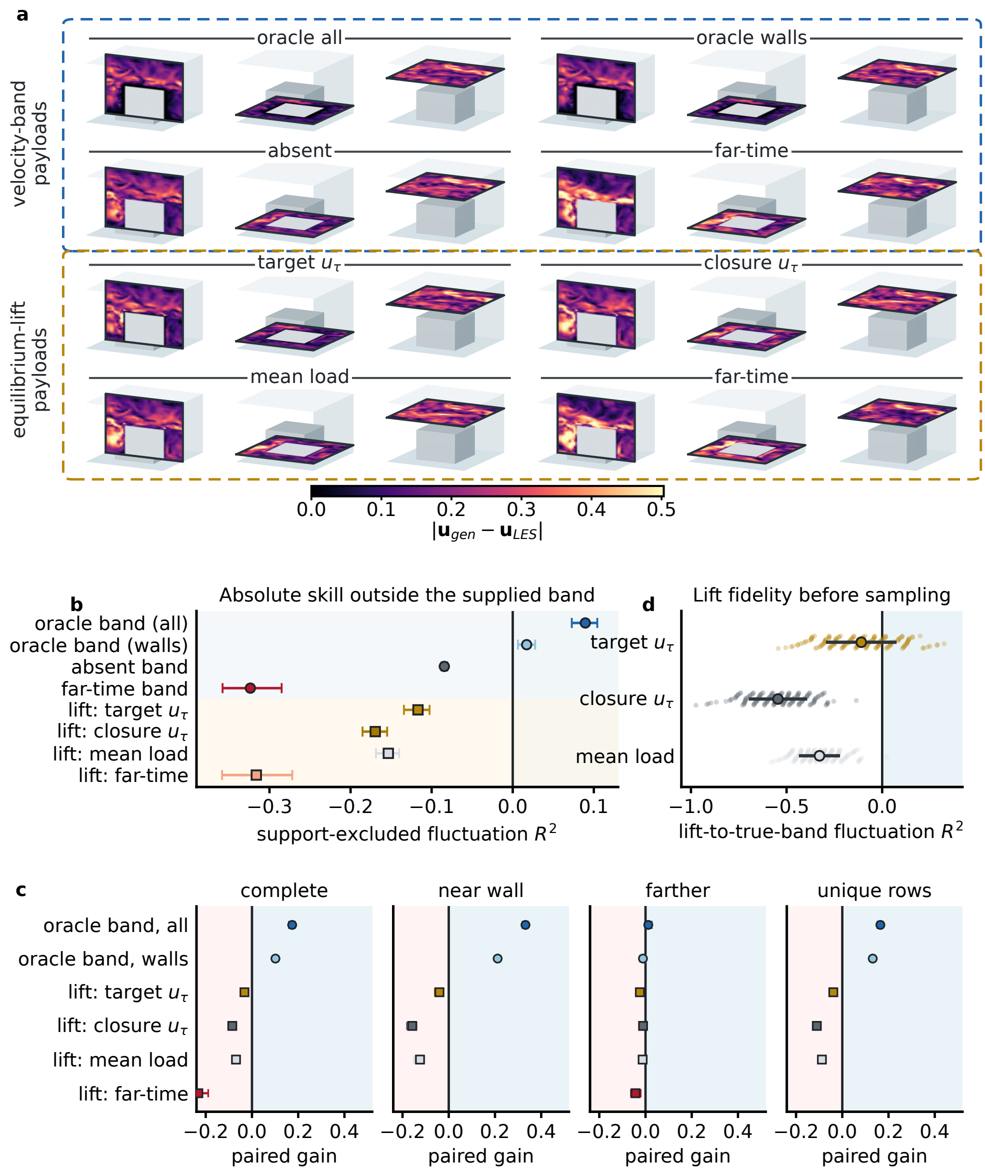

Fig. 4 Frozen closure-to-generator interface. Only payload changes. **a**, Representative 3-D errors for eight payloads on mid-span, near-wall and outer planes. **b**, Support-excluded R^2; a model-predicted traction with no target-derived wall information (friction-velocity field correlated 0.782 with the inversion) scores $-0.085\,[-0.098, -0.073]$, and the instantaneous inversion beats the mean wall load by $+0.037\,[0.021, 0.054]$. **c**, Gain over absence by region; whiskers in **b**,**c** are conservative 95% intervals. **d**, Target-wise lift fidelity (mean ± s.d.). Oracle bands help; every lifted-traction arm underperforms absence. This result covers only the frozen lift–generator composition, not general wall closures or solver-coupled performance.

2.3 A first-cell wall traction propagates without any velocity reconstruction

Removing the adapter requires conditioning on the surface quantity itself. The tangential projector annihilates the isotropic pressure term (Eq. (2)), so a wall-on-fluid traction is recoverable from the velocity record alone (Eq. (9)): a *first-cell viscous traction* on the model raster, not the native spectral-element derivative. Physical consistency checks ran before any arm was sampled (Methods). Generators were then trained from scratch to condition directly on that traction—no velocity reconstruction, no clamp—under a protocol frozen and hash-bound before the production run, which discloses prior contact: a previously contacted record, not an untouched confirmation ($N_{\text{eff}} = 2.46$).

The contrast is positive in every scoring region (Fig. 5). On the complete unsupplied volume the first-cell traction improves support-excluded fluctuation R^2 by $+0.074\,[0.070, 0.078]$ over the absent-conditioning branch, with the same sign on all three seeds (across-seed s.d. 7.52×10^{-4}); the near-wall gain is $+0.140\,[0.135, 0.145]$, decaying outward (Fig. 5e). Against a matched ceiling—the oracle band on the same support, retrained identically ($+0.117\,[0.111, 0.123]$)—the traction transmits $0.631\,[0.622, 0.639]$.

The controls place the effect in the wall information, not the act of conditioning: the traction arm beats a training-window mean load, a far-time, a shuffled and a sign-reversed traction by $+0.077$ to $+0.204$ on the complete volume (Fig. 5d). The shuffled and sign-reversed fields are out-of-training inputs measuring sensitivity under distribution shift, so the signed orientation is not by itself established as causal. One contrast runs the other way: farther from the wall the sign-reversed arm beats the correct one by $0.022\,[0.011, 0.032]$.

Three limits bound this. The traction is read from the held-out target, so it measures what a closure's output could transmit—closure accuracy is the question Sec. 2.4 answers. It is an invertible linear map of the first-cell tangential velocity (Eq. (9)), strictly less informative than the band, so a transmission ratio below one is not by itself evidence of a lossy interface. And absolute skill remains negative (-0.014 against -0.088 for absence).

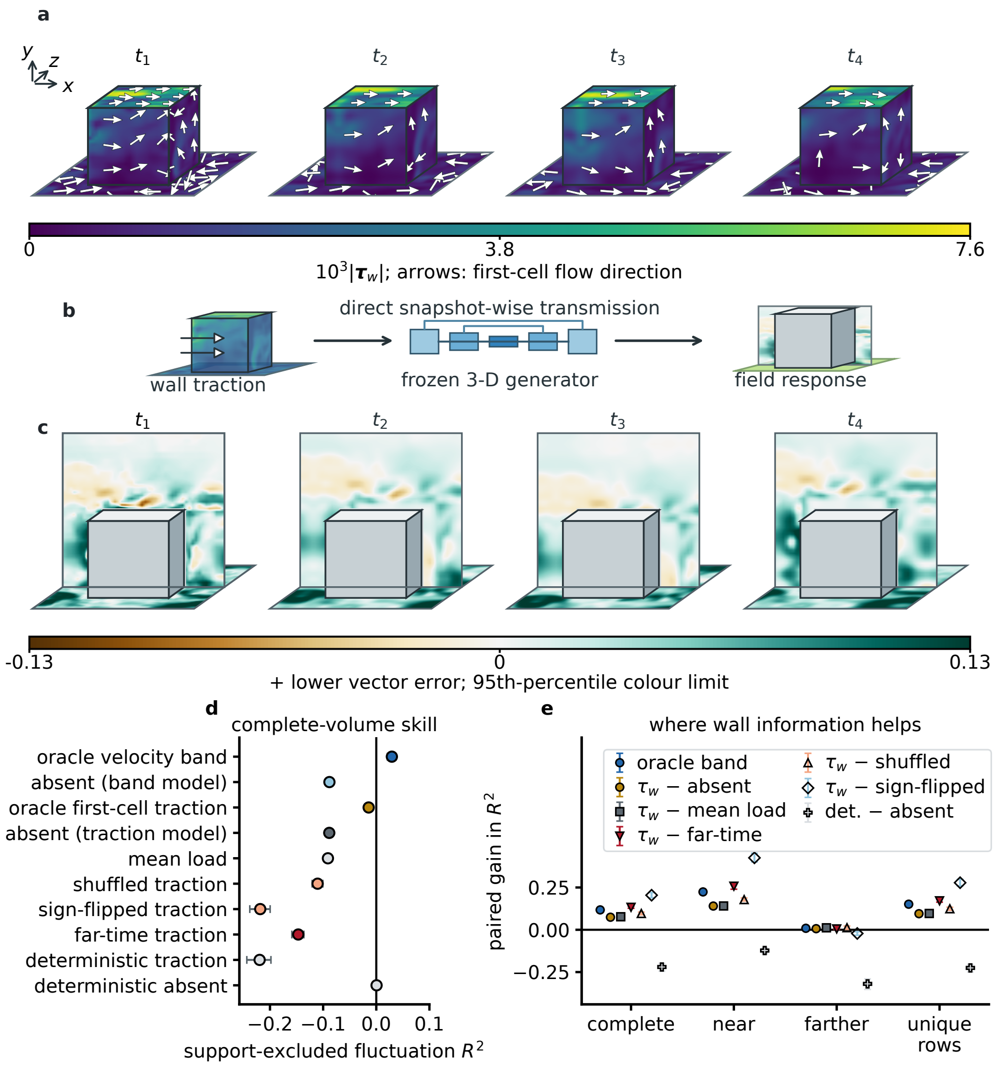

Fig. 5 Direct traction retains useful near-wall information. **a**, Wall-traction magnitude and first-cell flow direction at four illustrative instants. **b**, Snapshot-wise transmission through the frozen 3-D generator. **c**, Near-wall and midspan vector-error reduction (green indicates improvement). **d**, Complete-volume skill across controls. **e**, Paired gains across four regions; open markers denote out-of-training probes. Intervals use conservative moving blocks. The traction is oracle—an invertible linear map of the first-cell tangential velocity—and evaluated offline.

2.4 A closure that never sees the wall propagates into the field

Here we close the last map. A closure is fitted on the training window alone, frozen by hash, and used to *predict* held-out signed wall traction from matching-height outer-flow state only—velocity at 2.5Δ and 4.5Δ from each face, the wall-modelled-LES seam a coarse solver carries. It never reads the wall-adjacent cell (Sec. 4.6), and its prediction is all the generator receives. The score excludes every supplied cell *and* every closure-read cell (8,040 more), with the registered rule evaluated on 103 hash-bound held-out frames outside the direct-traction experiment's index, prospectively unscored with prior contact recorded (Supplementary Note 1).

The closure. On those frames the closure predicts the first-cell traction with zero-traction-reference skill 0.540 and mean direction cosine 0.746, against 0.413 for the classical equilibrium wall model it contains as the case $a = \theta = 0$ (Fig. 6e,f). The statistic normalises mean squared error (MSE) by the training-window root-mean-square rather than centring it (both conventions in Sec. 2.6); fit, validation and held-out scores are 0.553, 0.539 and 0.540, consistent with a 5,426-parameter correction that has not overfitted.

The composition. Conditioning the generator on that predicted traction improves source-excluded fluctuation R^2 by $+0.061\,[0.057, 0.064]$ over the absent-conditioning branch of the same network, with the same sign on all three seeds and a crossed seed×block interval of $[0.055, 0.065]$ that also excludes zero. The gain is $+0.118\,[0.112, 0.124]$ near the wall and positive under all seeds farther out (Fig. 6h). All five registered clauses pass, computed mechanically from the released decision-rule record. This is the paper's central claim: a wall quantity a closure actually produces—not one read from the answer—propagates into the surrounding field.

Controls. The decisive control is a far-time traction—a genuine traction of the wrong instant, and a training class of this generator, so no distribution-shift confound. It is *worse than supplying nothing* ($-0.043\,[-0.053, -0.030]$), and the closure beats it by $+0.104\,[0.092, 0.114]$: wall information helps only at the right instant. A mean-load, a shuffled and a sign-reversed traction are also below absence (Fig. 6h).

Where the gain comes from. At matched input information the learned non-equilibrium correction is worth $+0.013\,[0.003, 0.021]$ over the pure equilibrium closure, and the closure arm exceeds the oracle traction arm by $+0.013\,[0.003, 0.021]$—not prediction beating truth, since the arms carry *different* information and a closure's matching-height view is at least as useful for propagation as an exact wall derivative. Complete-volume absolute skill nevertheless remains negative for every arm (-0.028 closure against -0.088 absence, -0.041 oracle, -0.131 far-time).

Distributional validation. The closure arm has the lowest continuous ranked probability score (CRPS), per-cell energy score and rank-histogram reliability deviation of any arm at every ensemble size (Fig. 6d): the improvement is distributional, not an artefact of averaging.

A bounded engineering consequence. The streamwise viscous wall force inferred from the generated field—an offline reconstructed first-cell diagnostic excluding pressure and form drag, not the traction a solver would consume—falls in relative RMSE from 0.099 to 0.061 with the closure, its correlation with the true load rising from -0.178 to $+0.741$ (Fig. 6g). The equilibrium closure is *worse* than absence here despite a similar field-R^2 gain: the learned correction matters far more for wall load than for volume-averaged skill.

Transfer and non-regression. The frozen generator of Sec. 2.3, reused byte-for-byte, gains $+0.036\,[0.024, 0.046]$ from the predicted traction against $+0.052$ for the oracle traction it was trained on—about seven-tenths, the expected cost of feeding a predicted signal to a model trained on measured ones. Replaying the frozen checkpoints on that experiment's exact frames reproduces its published values within the frozen 0.005 identity tolerance, and all eight prospectively frozen tolerances pass.

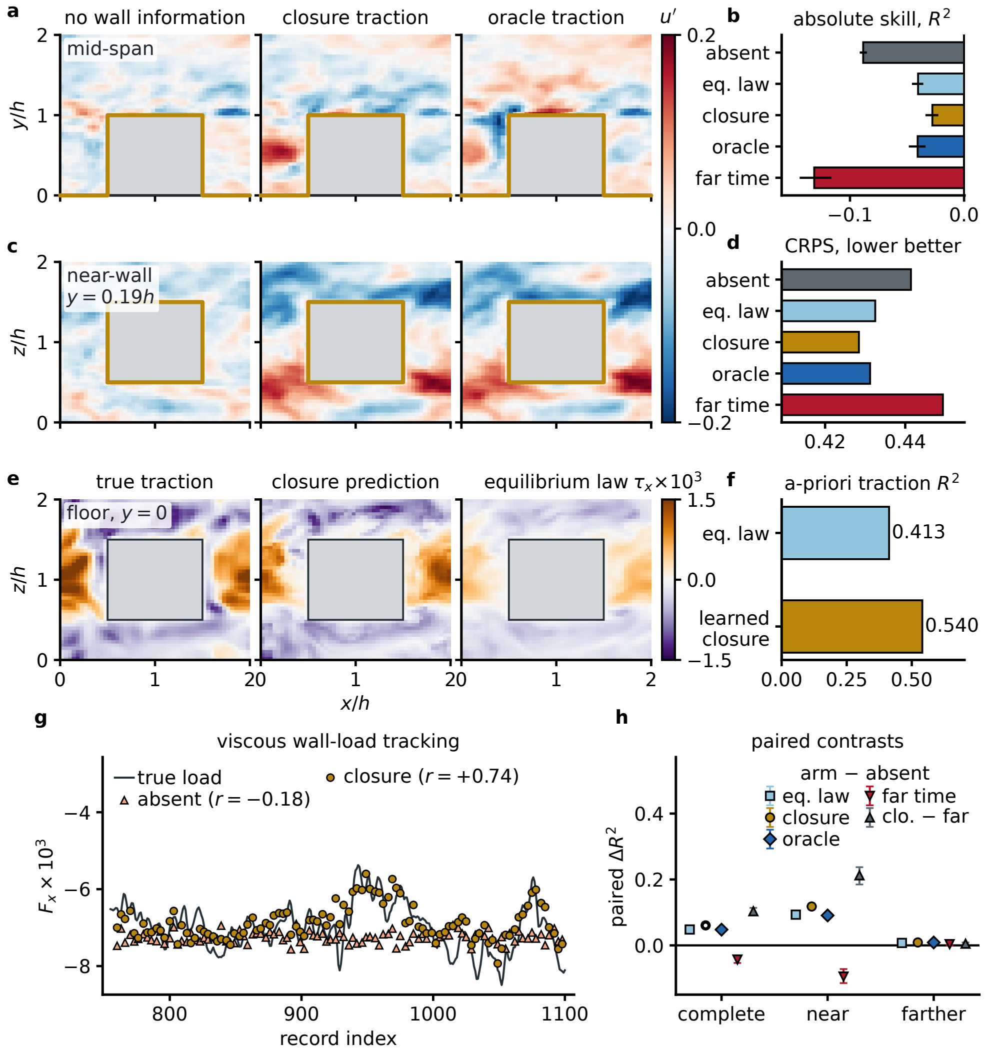

Fig. 6 A physics-grounded wall closure supplies traction that propagates through the generative model. **a**,**c**, Posterior-mean streamwise fluctuation of held-out record 760, mid-span and near-wall ($y = 0.19h$): absent, closure, oracle traction; gold marks traction-carrying walls. **b**,**d**, Absolute source-excluded skill (every arm negative) and CRPS; closure-arm energy score 0.922 against 0.948 (absent), reliability deviation 0.146 against 0.165. **e**, Floor traction: true, closure-predicted, equilibrium. **f**, A-priori traction skill. **g**, Implied viscous wall force vs true load (oracle: relative RMSE 0.035, correlation +0.927; equilibrium 0.160, bias +0.147). **h**, Registered paired contrasts; open circles, seeds. Scored on 103 prospectively unscored frames, outside every supplied and closure-read cell; offline.

2.5 The interface transmits on a separating flow

The cube wall is geometrically pinned. To ask whether the composition survives a wall whose stress is set by the flow, the identical protocol was re-executed on the *separating periodic-hill* flow [71–73] at $Re_h = 2800$ in both generative families and at two rasters, together with the cube under a *denoising-diffusion* generator [25]. The hill wall is smooth and curved, with pressure-gradient-induced separation—the regime where equilibrium wall models fail (Fig. 7a–c). It is a 2-D (u, v) corpus with no spanwise dynamics.

Two properties of the protocol travel to any wall-to-field study. *The evaluation unit is the physical time, not the stored array*: this corpus stores three spanwise planes per physical time, correlated up to 0.996, so scoring groups planes by physical time and separates training from scoring by 110 of them. A flattened array-index split instead hands each test frame its own same-time twins during training, inflating R^2_{fluct} by 0.839 (Supplementary Note 10). And *a unit must show it can detect wall information before it may adjudicate the interface*: no conclusion is drawn unless conditioning on true near-wall information improves reconstruction with an interval excluding zero. A contact audit shows no physical time of this record is untouched by earlier analyses, ours included.

The record passes that control in the diffusion family, and so does the method. Supplying the traction *directly*—the surface quantity the generator was trained to read—the record's own exact direct-numerical-simulation (DNS) traction improves source-excluded fluctuation R^2 by $+0.058\,[+0.009, +0.107]$, and the traction predicted by the frozen closure, which never reads a wall-adjacent cell, improves it by $+0.062\,[+0.017, +0.106]$, both proper scores agreeing (CRPS $-0.0124\,[-0.0188, -0.0056]$; energy score $-0.0172\,[-0.0268, -0.0076]$). Two paired contrasts say what that gain is made of: the closure arm is not resolved from the exact traction ($+0.004\,[-0.007, +0.017]$)—no advantage of the wall truth over the fitted closure is detected here; and the traction signal is *read* rather than merely present, the closure arm beating the same traction from a wrong instant by $+0.045\,[+0.001, +0.086]$ over the volume and $+0.101\,[+0.017, +0.170]$ near the wall, where wrong-instant information falls *below* supplying nothing.

The bounds matter as much. The near-wall interval for the closure arm alone crosses zero ($+0.063\,[-0.022, +0.152]$), and the arm is separated from neither a magnitude-matched scrambled traction ($+0.026\,[-0.021, +0.067]$) nor the equilibrium law ($+0.011\,[-0.015, +0.031]$), so specificity holds against wrong timing but is unresolved against wrong spatial pattern; absolute skill stays negative for every arm. With 370 scored times against a measured integral time of 112.3, $N_{\text{eff}} \simeq 3.3$: every hill interval here, favourable and adverse alike, is indicative rather than precise, and the hill is not part of this paper's headline claim.

How the traction is presented decides whether it transmits. Expanding it first into a near-wall velocity band through the closure's Reichardt profile does *not* clear zero at either raster ($+0.038\,[-0.017, +0.091]$ at 64×192; $+0.059\,[-0.011, +0.122]$ at 96×288), despite that lifted band reproducing the true band velocities at $R^2 = 0.871$ and 0.915 and the equilibrium lift matching it to three decimals: fidelity in the lifted representation is not sufficient. The oracle band arm, which *is* the record's exact band rather than a lift, does pass at both rasters ($+0.057\,[+0.011, +0.107]$ and $+0.079\,[+0.027, +0.131]$; Fig. 7f); its scrambled control is itself weakly positive over the whole volume ($+0.018$), so part of any whole-field gain is non-specific, but near the wall that vanishes and wrong information falls below absence.

At the wall the record is positive in both families. The closure predicts wall traction on the time-disjoint unit with $R^2 = 0.757$, against 0.557 for the equilibrium law it contains (Fig. 7d), and conditioning on it improves the reconstructed wall force ($0.036 \to 0.542$ and $-0.104 \to 0.445$ in correlation)—an offline reconstructed first-cell viscous-force diagnostic excluding pressure and form drag. The wall quantity is recovered even where the outer field is not: the near-wall/whole-field dissociation volume-averaged objectives conceal. The flow-matching cell fails the same positive control at both rasters, so by the standing rule it cannot adjudicate the interface in either direction; that arm and its failed repair are treated in the Discussion. Every registered contrast, sampler, raster and control is tabulated in Supplementary Note 10, with two further checks that fail their preregistered criteria, the withdrawn composition attempts and the custody audit that forced refitting the closure on the cube record (Supplementary Notes 6–8).

2.6 A re-test on a record that passes the control shows the interface transmits, and how far

To decide the whole-field question on a unit with demonstrated detectability, we re-executed the frozen closure→traction→generator composition on the wall-resolved cube record with three changes fixed before any outcome existed (Supplementary Note 11): the split is *reversed in time*, everything refitted from scratch on late frames and scored on a never-tested 25-frame window 132 frames from training; both families are trained at an identical budget with one frozen conditioning mixture including the oracle band, so band and traction arms are compared *within one propagator*; and the near-wall region $d \leq 0.5$ is co-primary with the whole field. That region is outcome-informed, hence evaluated on a fresh unit.

The positive control passes in both families ($+0.130\,[0.115, 0.145]$ and $+0.029\,[0.026, 0.032]$; Fig. 7g), and so does the interface. Conditioning on traction predicted by a closure that never reads a wall-adjacent cell raises near-wall fluctuation R^2 from -0.098 to $+0.005$, a gain of $+0.103\,[0.091, 0.115]$ (Fig. 7h)—the first arm here with positive absolute skill, recovering 79% of the DNS-band ceiling in the same propagator. The matched controls separate cleanly: a spatially shuffled traction of identical support and magnitude multiset scores -0.128 and a wrong-instant field -0.164, both *worse* than supplying nothing—the generator reads arrangement and instant, not the presence of a channel.

The effect has the wall-normal structure a boundary condition must have (Fig. 7i): in nested shells the same frozen contrast decays monotonically from $+0.266\,[0.221, 0.317]$ at $d \leq 0.15$ to $+0.006\,[0.004, 0.008]$ beyond $d > 0.5$, the nearest scorable shell reaching $R^2 = +0.149$ absolute against -0.117 unsupplied. The gain is 5.0 times larger beside the wall than in the whole-field average and 46 times larger than in the outer region, while the whole-field gain, $+0.053\,[0.046, 0.063]$, replicates the registered $+0.061$ of Sec. 2.4 on an independent, reversed-time unit.

Because the decisive window has $N_{\text{eff}} = 1.21$, the claim does not rest on the interval: the paired per-frame difference is positive in 25 of 25 held-out frames for both flow-matching seeds (minimum $+0.042$; diffusion 25/25 and 24/25; Fig. 7j).

The engineering-facing consequence is the reconstructed first-cell *viscous* wall force, an offline surrogate excluding pressure and form drag: closure conditioning moves its correlation with the true load from -0.14 to $+0.51$ (Fig. 7k; Supplementary Note 11). One adverse detail is reported: the shuffled control attains a *higher* wall-force correlation ($+0.61$), mechanically, since permuting whole traction vectors leaves the integrated force invariant—this integral cannot discriminate arrangement while the field endpoint does.

Two limits are registered rather than argued away. The rule frozen before this run returns **fail** on one clause—the diffusion family's *absolute* unconditional skill, -0.53, does not reach the inherited -0.14 floor, even though its interface delta is positive and interval-separated; a development sweep locates that in the family's posterior mean rather than in the interface, and the threshold was not adjusted after the outcome. And while the closure captures the held-out traction pattern better than the equilibrium law (Fig. 7e), the instantaneous departure from the held-out mean traction remains largely unpredicted (Supplementary Note 11).

As in Sec. 2.4 the closure arm slightly exceeds the *oracle traction* arm ($+0.005$ versus -0.015); the oracle band remains above both.

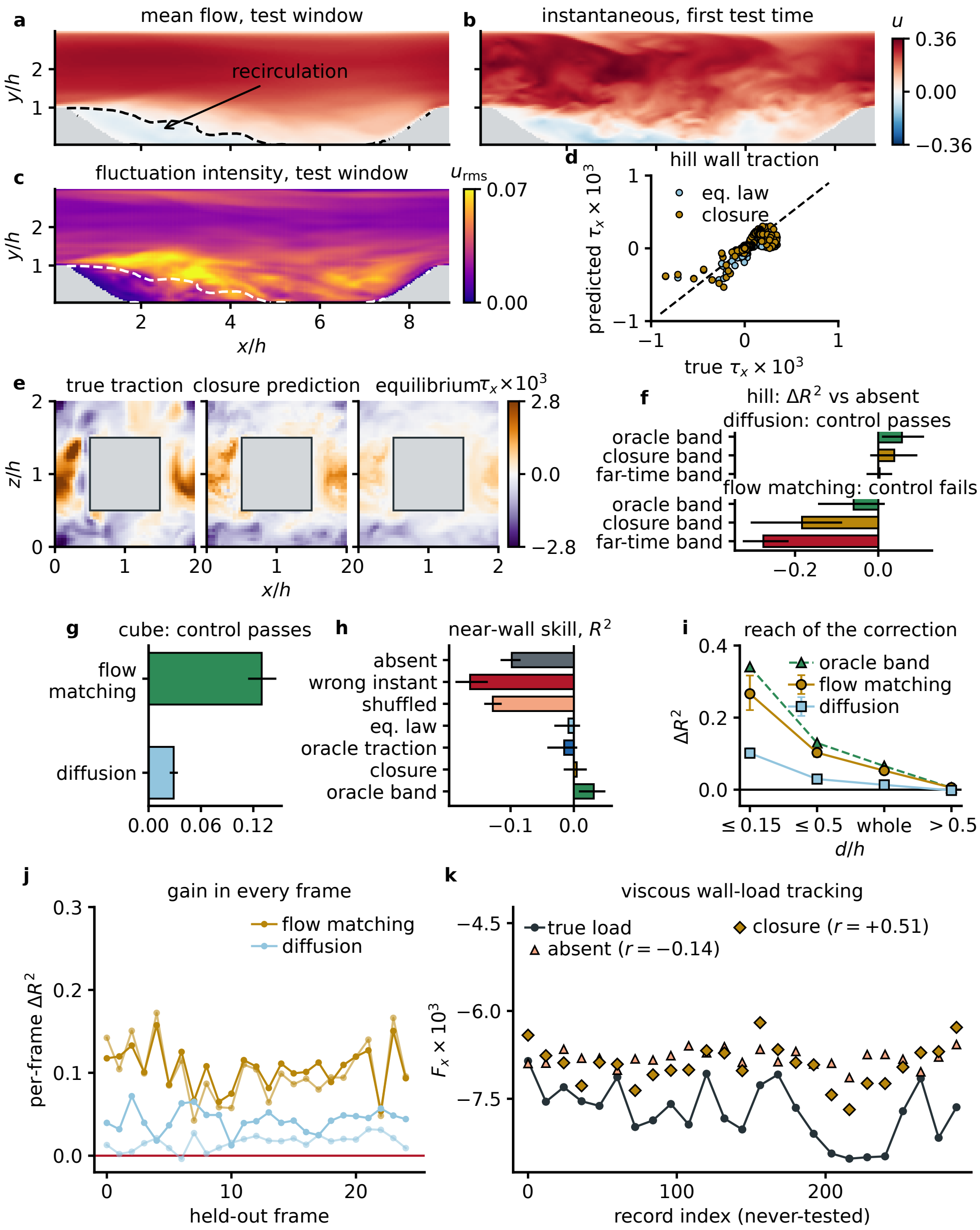

Fig. 7 A prospectively frozen oracle positive control decides which cells may adjudicate the interface. a–c, Separating hill: test-window mean flow (dashed, $\bar{u} = 0$), an instantaneous frame, shear-layer fluctuation intensity. **d**,**e**, Held-out traction on the separating wall and a never-tested cube frame. **f**, Each band arm's paired contrast against its own absent branch at 64×192; the label gives that family's *positive-control* verdict, not a verdict on the closure arm. The flow-matching failure survives a stochastic sampler and a finer raster (Supplementary Note 10). **g**, Positive control on the reversed-time cube. **h**–**k**, Near-wall ladder; reach against ceiling; per-frame gain; viscous wall-load tracking. All endpoints offline.

2.7 A single-slot channel experiment measures what transmitted traction is worth

Earlier generators saw named wall-signal classes during training. A prospectively registered single-slot experiment on a second, wall-resolved record—the smooth-wall channel of the public CoNFiLD release at $Re_\tau = 184$—removes that asymmetry. The generator is trained on exactly one traction-carrying condition, the record's own signed traction under a variance-preserving fidelity ladder, so every wall model enters the same slot with zero generator-side training exposure. A sublayer projection makes the generated wall gradient equal the supplied traction identically, the conditioning rows ($y^+ \leq 3.2$) are machine-verified absent from every scored cell, and the never-scored 46-frame window was contacted only after a two-stage preregistration froze every decision rule, control and margin (Methods).

On that window the record's own traction—the training modality—raises support-excluded fluctuation R^2 by +0.165 ($t = 204.7$ against a simulation-calibrated critical value of 4.5), and four graded reference arms trace a strictly monotone response (+0.104 to +0.153 at fidelities 0.082–0.797; Spearman 1.0; Fig. 8g). The closure, reading only two matching-height velocity lines and never any cell below $y^+ \approx 29$, gains +0.120 over supplying nothing ($t = 36.5$; every block, seed, wall and frame individually positive), with *positive* absolute skill (+0.030 and +0.035 by seed against −0.091 and −0.083) and the wall-normal structure a boundary condition must have: +0.328 in the buffer layer, +0.066 at $30 < y^+ \leq 100$, +0.0001 beyond (Fig. 8f).

Attribution is read from the registered controls. The closure stays below the exact-traction reference (−0.045) and, at its own fidelity (0.023), sits +0.016 above the lowest rung, inside the frozen 0.02 margin. The data-only regressor, with better fidelity (0.083), gains more (+0.131): scalar fidelity orders, but does not alone predict, downstream gain, and the ladder is a within-family sensitivity, not a universal law. Wrong wall information does not merely fail: a wrong-instant traction scores −0.060 *below* supplying nothing (closure better by +0.180), a spanwise-shuffled traction is beaten by +0.144 and the equilibrium law by +0.044. The mechanically applied rule concludes the interface transmits traction information within the frozen margins.

The registered engineering endpoint improves: in the buffer band, outside every conditioning and read row, the tracking error of the spanwise-averaged turbulent shear $\langle u'v' \rangle$ falls by 26% with the closure ($0.191 \to 0.141\, u_\tau^2$; $t = 8.61$) and 37% with the exact traction. CRPS improves in the same direction ($t = 98.7$; secondary); the pooled rank histograms [74] are U-shaped for every arm, an underdispersion reported descriptively. The diffusion family reproduces the primary contrast with both seeds positive (+0.106).

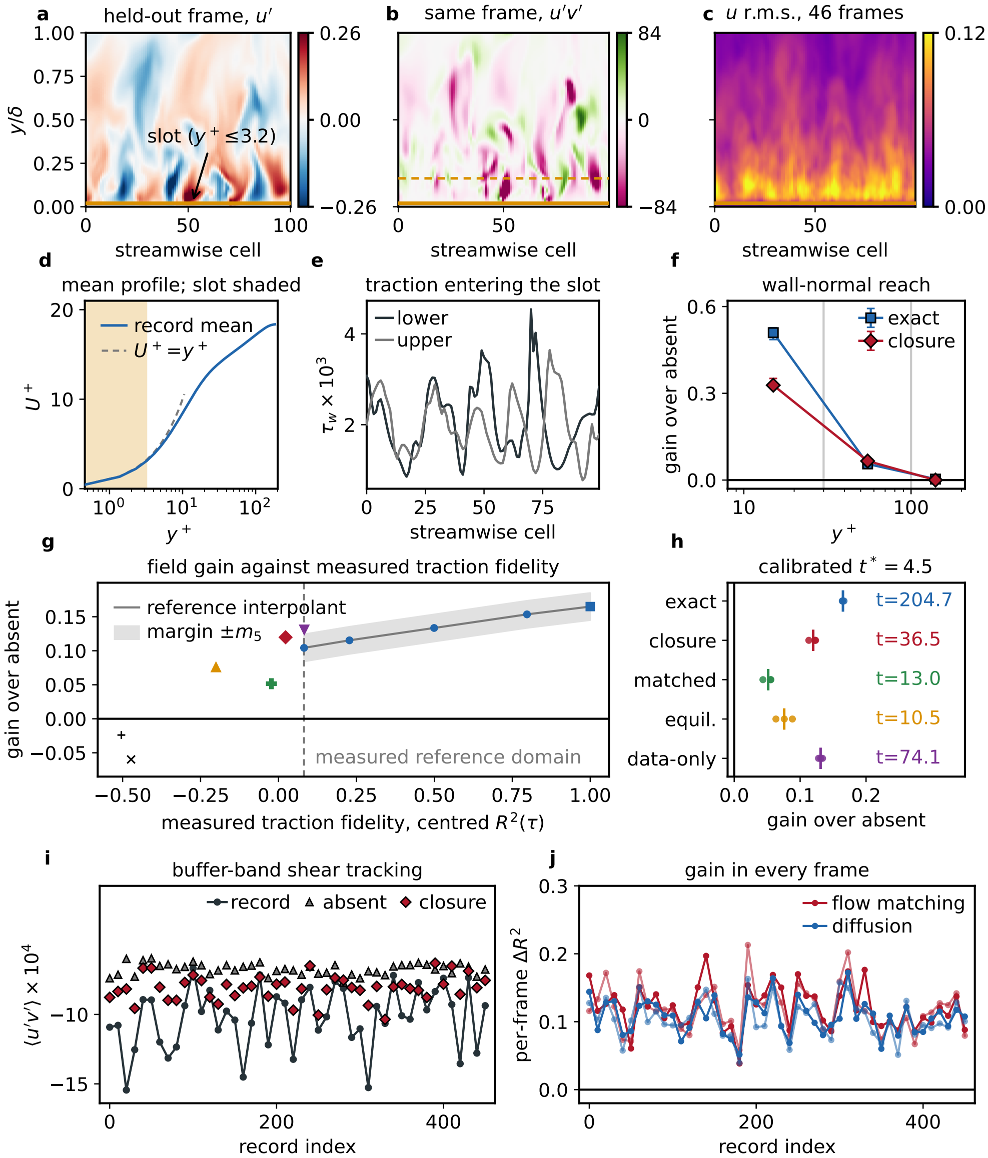

Fig. 8 A single-slot channel experiment: transmitted traction information improves the generated field. **a**–**c**, The wall-resolved record: a held-out frame's u', its $u'v'$ (buffer band dashed), and u r.m.s. over 46 frames; gold marks the conditioning slot ($y^+ \leq 3.2$). **d**,**e**, Mean profile in wall units; slot traction on both walls. **f**, Wall-normal reach. **g**, Field gain against measured traction fidelity; the lowest rung ($r = 0.55$) still correlates structurally with the true traction. **h**, Block inference at the calibrated $t^* = 4.5$. **i**, Reconstructed buffer-band turbulent shear tracking: per-frame error 3.84×10^{-4} (absent), 2.83×10^{-4} (closure), 2.40×10^{-4} (exact traction). **j**, Per-frame gain; every frame, seed and family positive.

3 Discussion

This paper evaluates one integrated system end to end: a physics-grounded wall closure connected to a conditional generator. Reading matching-height flow state, the closure predicts held-out traction better than its equilibrium law (0.540 against 0.413); supplying it improves source-excluded fluctuation R^2 by $+0.061\,[0.057, 0.064]$ under a pre-fixed rule. The unit was prospectively unscored, not independent.

On the separating hill, the diffusion positive control passes and closure traction gains $+0.062\,[+0.017, +0.106]$, not resolved from exact traction and better than wrong-time traction. Roughly three effective samples make this indicative rather than precise.

One preregistered arm was adverse: flow matching fails its positive control and cannot adjudicate the interface. We tested the obvious under-dispersion explanation prospectively with a stochastic sampler on byte-identical weights; dispersion improved while the control failed *more* clearly. A model-family difference in how sparse wall information is used remains plausible, but no mechanism is established. Ensemble dispersion is therefore reported as a diagnostic, never an acceptance criterion (Supplementary Note 10).

Two properties make the cube number interpretable: its decisive control is a real traction of the wrong instant, scoring worse than supplying nothing, so the model is rewarded for *this instant's* wall state; and the gain is distributional, the closure arm best on every proper score at every ensemble size.

The practical reading is specific: expanding wall stress into an equilibrium-law velocity band and clamping it fails, whereas the same quantity transmits through conditioning trained to read it. The instantaneous field a generator needs is not the field an equilibrium law describes.

The limits are stated once, each marking where the evidence ends and the next study begins. The composition is established offline and a priori, never inside a time-evolving solver. Complete-volume absolute skill on the cube record stays negative for every arm (-0.028 closure against -0.088 absence)—wall information moves the field towards the target without yet making it accurate—while the re-test's near-wall region and the channel record reach positive absolute skill. All three records carry few effective independent events, so every interval is conditional on them—the price of difficult engineering records over abundant homogeneous ones. The farther-from-wall gain, $+0.009\,[0.003, 0.016]$, is the honest measure of reach; the out-of-training probes are sensitivity checks, not causal controls; and the fitted target is a first-cell viscous traction on the model raster, not the native spectral-element gradient. The closure reads its matching heights at fixed multiples of the local cell size, 2.5Δ and 4.5Δ—the seam a wall-modelled solver carries—so the multiples and the mesh setting Δ are parameters of the interface itself, and the natural first task for a deployment study is charting the gain's response to coarser rasters and different seams. The same standard then extends to the regimes most likely to strain the interface: unresolved near-wall bands, spanwise three-dimensional near-wall structure, rough walls and Reynolds numbers far above these records, where the wall's far-field footprint grows.

The supportable engineering consequence is offline and specific: the implied streamwise viscous wall-force error falls by almost two-fifths with the closure, while the equilibrium closure is worse than absence on this endpoint despite a similar field-R^2 gain—a wall-load criterion discriminates between closures that a volume-averaged score does not. Additional closure-side geometry-transfer, separation, two-to-three-dimensional-transfer and diversity-scaling strengths come from separate frozen companion models preserved by unchanged source hashes; the present generator compositions do not revalidate those claims.

Wall-observation studies show that deployable pressure and shear signals can inform turbulent fields [51, 52, 57], and field-generative studies demonstrate wall-bounded generation, super-resolution and sparse reconstruction [43–45]. This work complements them by taking the wall quantity from a closure rather than an observation, excluding supplied and read cells from every score, and testing against an in-distribution wrong-information control. It establishes no superiority over prior architectures: no head-to-head comparison was run.

What a reader can take from this is separable from our own closure. The conditioning slot is a harness, not a result: a wall model supplying a signed two-component wall-on-fluid traction enters as a surface field on the wall faces alone, the generator never sees which model occupies it, and a sublayer projection makes the generated wall gradient equal the supplied traction identically, so two wall models compete with no generator-side advantage (Fig. 8). The fidelity–gain response measured through it, monotone from +0.104 to +0.153 across traction fidelities 0.082–0.797, states how accurate a wall model must become before a generative reconstruction benefits; its limit is equally usable, since a data-only regressor of better scalar fidelity gains more than the ladder predicts, so error *structure* matters beyond any scalar. We propose the oracle positive control as the minimum adequacy standard such a study needs before reading a conditioning contrast, and the grouped physical-time protocol, with the 0.839 inflation a naive flattened split produces, applies to anyone evaluating this widely used public corpus. All are released with the code and data.

The three records show that generative models do not bypass the wall problem: they demonstrate the framework on pinned, attached and separating cases, not a claim to have covered wall modelling. What remains is runtime composition: a solver-coupled calculation supplying a momentum boundary condition as the flow evolves. Transmission holds in both generative families on the pinned cube and attached channel records, and in one of the two on the separating record, the other an adverse preregistered arm for which no mechanism is claimed. Within those bounds a physics-grounded wall closure can supply information a generative model demonstrably propagates through the field.

4 Methods

4.1 Evidence classes, data contact and estimands

We distinguish three evidence classes. The first, *oracle wall-band propagation*, supplies instantaneous large-eddy-simulation (LES) velocity to the generator and excludes that support from every score. The second, *closure-supplyable wall information*, splits into $L_A : \mathbf{q}^{\text{coarse}} \to \boldsymbol{\tau}_w$, the closure's own accuracy, and $L_B : \boldsymbol{\tau}_w \to \mathbf{o} \to p_\theta$, the propagation of a wall traction into the field. The third, *closure-side transfer*, comprises companion evidence from separate frozen models.

Oracle-band propagation, L_B and the composition $L_A \circ L_B$ are evaluated offline. L_B is evaluated twice and independently: through a frozen equilibrium velocity reconstruction and by conditioning a generator directly on the native signed wall traction of Eq. (9) (Sec. 4.5). Both use an *oracle* traction read from the held-out target, so they bound what a closure could transmit and are not closure-accuracy measurements. L_A is closed in Sec. 4.6: a closure is fitted on the training window, frozen by hash, and used to predict held-out traction from matching-height state alone, so the composed chain is evaluated end to end. Earlier candidates for L_A are excluded and retained as labelled diagnostics (Sec. 4.10); nothing in this work advances a momentum equation.

Chronological separation does not make the oracle-band record untouched. M0 encountered the evaluation interval after allocation and arm definitions were frozen; M1 then changed capacity, optimisation and sampling, and M2 changed padding, all on the same targets; the ensemble endpoint was frozen after M2 selection but before its inference. We call these the *first-contact comparison*, *model-development sensitivity* and *post-selection registered endpoint*, and none is called an independent confirmatory test.

Point estimands are absolute fluctuation R^2 for the aligned band and aligned-minus-control differences over complete, near and geometrically farther $(d/h > 0.5)$ support-excluded regions. The registered ensemble estimand is farther-region energy score.

Statistical and scope conventions, used throughout the paper. Every evaluation is offline and a priori. Reported intervals are conservative moving-block resampling intervals, conditional on the finite record and fitted model, and not multiplicity-adjusted; evaluated frames are temporally correlated rather than independent events, with effective event counts stated where decisive (the cube record's 343-output span is 2.800 conservative integral times), and each generative point estimate is one realised eight-sample mean. The absent arm is always a conditioning-dropout branch of the same network, not a budget-matched unconditional generator, so each paired gain is a matched information intervention inside one conditional model. M0 retains only terminal aggregates. No evaluation unit in this work is claimed to be untouched: units are prospectively unscored, with all prior contact recorded in a global contact ledger (Supplementary Note 1).

4.2 Wall-to-field interface and evidence boundary

The intended integrated map is

$$\mathbf{q}_t^{\text{coarse}} \xrightarrow{\mathcal{H}} (\mathbf{\Phi}_t, \mathbf{S}_t) \xrightarrow{\mathcal{C}} \boldsymbol{\tau}_{w,t}^{\mathrm{W}\to\mathrm{F}} \xrightarrow{\mathcal{L}} \mathbf{o}_t \xrightarrow{\mathcal{G}} p_\theta(\mathbf{u}_t \mid \mathbf{o}_t), \tag{1}$$

where $\mathcal{H}$ constructs the local rows $\mathbf{\Phi}_t$ and family-level station set $\mathbf{S}_t$, $\mathcal{C}$ is the closure, $\boldsymbol{\tau}_w^{\mathrm{W}\to\mathrm{F}}$ is wall-on-fluid traction, $\mathcal{L}$ lifts it to the generator representation and $\mathcal{G}$ samples the field. A causal claim additionally requires runtime-available or prospectively calibrated inputs, an executed lift, untouched evaluation targets and information-matched controls.

The frozen companion closure does not supply such a general runtime map: its inference header stores a fixed 16-dimensional branch embedding exported from the Krank $Re = 10{,}595$ periodic-hill member's own wall-stress-free station set [75], without the branch weights or set normalisation needed to recompute it for a new source, so the custody of $\mathbf{S}_t$ and the calibration-consistent pressure-gradient and displacement-thickness features remain unresolved closure-composition dependencies (Sec. 4.10).

For the present nonplanar surface the required closure output is a tangential vector, not one signed global scalar. Let the unit wall normal $\mathbf{n}$ point from the solid into the fluid. The kinematic traction exerted by the wall on the fluid is

$$\mathbf{P}_t = \mathbf{I} - \mathbf{n}\mathbf{n}^\mathsf{T}, \qquad \boldsymbol{\tau}_w^{\mathrm{kin,\,W\to F}} = -\mathbf{P}_t\big[\nu(\nabla\mathbf{u} + \nabla\mathbf{u}^\mathsf{T})\mathbf{n}\big] = \tau_1^{\mathrm{W\to F}}\mathbf{t}_1 + \tau_2^{\mathrm{W\to F}}\mathbf{t}_2, \quad (2)$$

with the fluid-on-wall traction of opposite sign. For a lower wall with $\mathbf{n} = \mathbf{e}_y$, $\mathbf{t}_1 = \mathbf{e}_x$ and $\partial u/\partial y > 0$, Eq. (2) gives $\tau_1^{\mathrm{W\to F}} = -\nu\,\partial u/\partial y < 0$. Here $\mathbf{t}_1$ is a geometry-anchored unit tangent fixed independently of the instantaneous flow and $\mathbf{t}_2 = \mathbf{n} \times \mathbf{t}_1$, declared face by face, so flow reversal changes the signed components in this fixed basis. A valid $\mathcal{L}$ must carry both signed tangential components, normals, local bases, volumetric support and the stress scaling.

Oracle-band propagation starts at the final map, $\mathbf{o}_t = \mathbf{1}_B \odot \mathbf{u}_t^{\mathrm{LES}}$ supplied by the target LES itself.

4.3 Aligned-cube simulation, masks and allocation

The aligned-cube record was generated with NekRS, a graphics-processing-unit (GPU) spectral-element incompressible Navier–Stokes solver [76]. The $2h \times 4h \times 2h$ domain contains one $h \times h \times h$ wall-mounted cube, giving pitch $2h$, plan solidity $\lambda_p = 0.25$ and $Re_h = \overline{U}h/\nu = 5000$; streamwise and spanwise boundaries are periodic, the floor, cube faces and ceiling no-slip, and a constant-flow-rate controller holds $\overline{U} = 1$. The mesh has 14,176 hexahedra; order-7 Gauss–Lobatto–Legendre (GLL) discretisation gives 7,258,112 points with order-9 cubature, two operator-integration-factor-splitting (OIFS) advection substeps at a target Courant–Friedrichs–Lewy (CFL) number of 2, maximum $\Delta t = 1.5 \times 10^{-3} h/\overline{U}$, pressure/velocity tolerances $10^{-4}/10^{-6}$ and one-mode high-pass filtering (coefficient 10) without explicit subgrid viscosity. The run ended normally at $t\overline{U}/h = 370.0005$ with 1,234 fields at spacing 0.3; after discarding $t < 40.2$, volume-mean kinetic-energy drift between the middle and final halves is 1.55×10^{-3}.

Each GLL field is mapped to a $96 \times 192 \times 96$ Cartesian raster by nearest-GLL lookup and exactly averaged over non-overlapping $2 \times 2 \times 2$ cells to the $48 \times 96 \times 48$ model grid. The fluid mask F contains 207,360 cells. The condition mask $B \subset F$ comprises fluid points within $2.01(4/96)h = 0.08375h$ of the floor, ceiling, cube tops or vertical faces—the nominal two-cell band of thickness $2(4/96)h = 0.08333h$—and contains 14,008 cells. We define

$$U = F \setminus B, \quad U_{\mathrm{near}} = U \cap \{d/h \le 0.5\}, \quad U_{\mathrm{far}} = U \cap \{d/h > 0.5\}, \qquad (3)$$

with $|U| = 193{,}352$, $|U_{\mathrm{near}}| = 74{,}844$ and $|U_{\mathrm{far}}| = 118{,}508$; the latter two are disjoint and partition U.

The nearest-GLL lookup does not preserve full wall-normal resolution in the upper domain. Comparing adjacent Cartesian rows over nine snapshots identifies exactly 30 row pairs, all at $y/h \gtrsim 1.7$, with bitwise-identical fluid values, and one ceiling-adjacent row (index 95) identically zero at every retained time: 34.56% of U lies on rows that repeat the row below. All 4,608 ceiling cells of the supplied band are affected, 2,304 identically zero and 2,304 bitwise identical to the first scored row beneath them—a leak handing the generator the exact values of 1.19% of U. Restricting the band to the physical walls removes both defects, and Sec. 2.2 reports every arm under that restriction; a fourth scoring region U_{uniq}, the 126,536 support-excluded cells on raster-unique rows, is reported alongside the three published regions. The paired interventions share the same mask and remain valid, but the farther-from-wall region is partly an interpolation artefact.

A native first-cell audit differentiates unrasterised GLL wall-face velocity on separate continuation fields at $t\overline{U}/h = 390.0003$ and 420.0001 (not independent replicates): maximum y^+ is 0.31631 on the exposed floor, 1.33684 on cube tops and 1.92126 on vertical faces. The ceiling was not classified, so the ceiling-containing band supports no ceiling-specific native y^+ claim.

After spin-up, 1,101 fields remain. Before any arm output, the feasible allocation was frozen as 660 training fields, a 98-field exclusion gap and a 343-field chronological remainder, whose first index 758 lies 99 output spacings after the last training index 659. Evaluation selects 160 evenly spaced fields. No separate validation interval exists, so later use of the 160 fields for model development is disclosed explicitly. Training-only statistics determine channel normalisation and the spatial mean used for fluctuations.

4.4 Conditional flow models, sampler and intervention arms

Let $\mathbf{x}_0 \in \mathbb{R}^{3\times N_x \times N_y \times N_z}$ be a standardised training field and $\mathbf{z} \sim \mathcal{N}(0, I)$ a Gaussian endpoint on fluid cells. For $t \sim \mathcal{U}(0,1)$, the rectified path and target velocity are [28, 77]

$$\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{z}, \qquad \mathbf{v}^\star = \mathbf{z} - \mathbf{x}_0. \tag{4}$$

The network input is $[\,\mathbf{x}_t,\ \mathbf{o},\ \mathbf{1}_B,\ \mathbf{1}_F\,]$, where $\mathbf{o} = \mathbf{1}_B \odot \mathbf{x}_0$ is a three-channel wall-band observation, and the fluid-masked flow-matching loss is

$$\mathcal{L}(\theta) = \mathbb{E}\left[\frac{\sum_i \mathbf{1}_F(i)\,\|\mathbf{v}_\theta(\mathbf{x}_t, t, \mathbf{o}, B, F)_i - (\mathbf{z} - \mathbf{x}_0)_i\|_2^2}{3|F|}\right]. \tag{5}$$

With probability 0.25, $\mathbf{o}$ and $\mathbf{1}_B$ are set to zero, training the absent-band branch; the far-time condition is not a training class.

All three sequential models are two-level three-dimensional U-Nets with sinusoidal time embeddings and group normalisation, trained by AdamW with learning rate 2×10^{-4}, betas $(0.9, 0.99)$, weight decay 10^{-4}, batch size 2, gradient clip 2 and exponential moving average (EMA) decay 0.9995. M0 has width 24, 2,213,139 parameters, 30,000 updates and 16 sampler steps; M1 width 48, 8,647,587 parameters, 90,000 updates and 32 steps; M2 keeps M1's budgets but uses circular streamwise/spanwise padding. Each configuration has one fit and eight samples per target. Training and inference

Algorithm 1 Matched wall-information intervention

Require: Chronological evaluation fields, masks B, F, fitted network and frozen sampler seeds

```
Define common score mask U ← F \ B and load training-only normalisation
for each evaluation field q do
    Set (o_aligned, B_aligned) ← (x_{q,B}, B), (o_absent, B_absent) ← (0, ∅) and (o_far, B_far) ← (x_{d(q),B}, B)
    for each stochastic sample m ∈ {1, ..., 8} do
        Draw one z_{q,m} and reuse it in all three arms
        for arm in {aligned, absent, far-time} do
            Integrate Eq. (6) with (o_a, B_a) for the model-specific steps
            Retain the generated field only on U
        end for
    end for
end for
Resample shared physical-time blocks and recompute absolute scores and paired contrasts
```

are deterministically seeded, with frozen inference seeds reused across matched arms; exact bitwise retraining is not claimed (Code availability).

Sampling integrates the learned velocity from $t = 1$ to $t = 0$ by explicit Euler updates. For arm a with conditioning $c_a = [\mathbf{o}_a, \mathbf{1}_{B_a}, \mathbf{1}_F]$, a supplied band follows the exact rectified bridge,

$$\mathbf{x}_{t_{j+1}} \leftarrow (1 - \mathbf{1}_{B_a}) \odot \left[\mathbf{x}_{t_j} + \Delta t\, \mathbf{v}_\theta(\mathbf{x}_{t_j}, t_j, c_a)\right] + \mathbf{1}_{B_a} \odot \left[(1 - t_{j+1})\mathbf{o}_a + t_{j+1}\mathbf{z}_{B_a}\right]. \quad (6)$$

The aligned arm uses $\mathbf{o} = \mathbf{x}_{q,B}$ with $B_a = B$; the absent arm uses zero observation and $B_a = \varnothing$ and is not clamped; the far-time arm uses $\mathbf{x}_{d(q),B}$, where $d(q)$ is a frozen circular shift by 80 evaluation positions (paired indices 172 or 173 native snapshots apart, approximately $1.40T_{\text{int}}$), preserving support while changing temporal alignment. All arms are scored on the same mask $U = F \setminus B$, and checkpoint, targets, integrator and sampler noise are identical across arms within each model (Algorithm 1).

4.5 Traction interface: equilibrium lift and direct conditioning

The load-bearing approximation in Eq. (1) is the composition $\mathcal{L} \circ \mathcal{C}$, separated into $L_A : \mathbf{q}^{\text{coarse}} \to \boldsymbol{\tau}_w$ and $L_B : \boldsymbol{\tau}_w \to \mathbf{o} \to p_\theta$. Because the frozen companion closure cannot be executed source-generally on this record (Sec. 4.2), L_A must be fitted on this record, which Sec. 4.6 does. Two earlier overgeneralisations do not hold: that "L_A cannot be executed" at all, and that a failed adapter bounds every composition from above—a failure of one adapter constrains that adapter, not the class. Three executions are reported in the order run—an equilibrium velocity reconstruction, direct traction conditioning, and the closure composition—and each protocol was registered before any of its arms was sampled.

Wall surfaces and anchors. The model grid resolves seven no-slip surfaces: floor, lid, cube top and the four cube vertical faces. Six are *physical* walls; the lid is a computational closure whose adjacent raster row is identically zero (Sec. 4.3), so it carries no wall observation. Each supplied-band cell is assigned to the surface of smallest perpendicular distance among surfaces whose in-plane extent, dilated by two cells, contains it; ties follow a fixed surface order. The anchor of a band cell is the first fluid cell of its owning surface, at perpendicular distance $\Delta/2 = 0.02083h$.

Equilibrium traction estimator and lift. With $\mathbf{u}_a$ the anchor velocity, $u_a = |\mathbf{P}_t\mathbf{u}_a|$ and $\hat{\mathbf{t}} = \mathbf{P}_t\mathbf{u}_a/u_a$, the friction velocity solves $u_a = u_\tau f_R(d_a u_\tau/\nu)$ by bisection, with the Reichardt profile

$$f_R(y^+) = \kappa^{-1}\ln(1+\kappa y^+) + C\left[1 - e^{-y^+/11} - (y^+/11)\,e^{-y^+/3}\right], \tag{7}$$

$\kappa = 0.41$, $C = 7.8$ and $\nu = 2\times10^{-4}$. Anchor y^+ spans 0.169–11.58 (mean 3.80), operating in the viscous and buffer layers. The wall-on-fluid traction is $\boldsymbol{\tau}_w^{\mathrm{W}\to\mathrm{F}} = -u_\tau^2\hat{\mathbf{t}}$. Because $\mathbf{u}_a$ is read from the evaluation target, this is a *target-velocity equilibrium inversion*, not native traction (Eq. (9)). Every band cell at wall distance d from its owning surface receives

$$\mathcal{L}(\boldsymbol{\tau}_w)(d) = u_\tau f_R(d\,u_\tau/\nu)\,\hat{\mathbf{t}}, \tag{8}$$

with zero wall-normal component; κ, C and ν were fixed before any field score was computed. Because no equilibrium observation exists at the lid, all traction arms condition on the physical-wall support—9,400 of the 14,008 supplied cells—and an additional oracle arm supplies the LES band on exactly that support, a ceiling matched to the same information geometry; scoring masks are unchanged. Three arms reproduce the published aligned/absent/far-time comparison as a non-fabrication guard. Four arms are new: the target-velocity inversion through Eq. (8); far-time traction through the same lift; the mean wall load, from the training-split mean field, carrying the same lift but no instantaneous wall information; and a model-predicted traction from the absent-band posterior mean's own anchor velocities, containing no target-derived wall information at any point. The interface is also scored a priori by the fluctuation R^2 of the lifted band against the true band on the physical-wall support.

Direct native-traction conditioning. The direct experiment removes the adapter and its two confounds (Sec. 2.2). The tangential projector in Eq. (2) annihilates the isotropic pressure term identically, so for each fluid cell adjacent to a no-slip surface, with $d = \Delta/2$, no-slip gives the one-sided estimate

$$\boldsymbol{\tau}_w^{\mathrm{W}\to\mathrm{F}} \simeq -\frac{\nu}{d}\,\mathbf{u}_t, \qquad \mathbf{u}_t = \mathbf{u} - (\mathbf{u}\cdot\mathbf{n})\mathbf{n}. \tag{9}$$

Because the raster velocity came from nearest-node lookup and $2\times2\times2$ averaging, this is a *first-cell viscous traction*, not the native spectral-element wall gradient; we use that name throughout. Two physical-consistency checks ran before any arm was sampled: the signed convention $\boldsymbol{\tau}_w^{\mathrm{W}\to\mathrm{F}}\cdot\mathbf{u}_t < 0$ is asserted as fatal on every face, and the implied streamwise wall-on-fluid viscous force (-0.00766 with each cell–face pair counted once) is compared against an independent spectral-element quadrature of the same stationary flow (-0.00742 and -0.00639; continuation-block audits, physical walls only)—an integral time-mean check, not a local instantaneous validation. Each (cell, face) pair contributes to the force exactly once; an earlier double count of the 96 two-face cells is corrected. Traction is carried by the six physical faces (4,512 cells), the lid again excluded, and cells touching two faces sum the per-face tractions.

Three generative models are trained prospectively from scratch under one budget: base width 32, 20,000 updates, batch 4, three independent seeds. Family T conditions on the traction of Eq. (9); family B conditions on the oracle velocity band restricted to the same physical walls (the matched ceiling); an identical architecture trained as a deterministic regressor gives an information-matched non-generative baseline. A 25% conditioning-dropout schedule means one network serves both the conditioned and absent arms of each family, so the primary contrast carries no between-model confound. The traction arms use *no* clamp: the surface information enters only through conditioning channels. Controls on family T are the training-window mean traction, a far-time traction, a sign-reversed traction and a traction shuffled across wall cells, all on identical targets, support and sampler noise; traction is normalised by training-window statistics only. The chronological split, exclusion gap and bootstrap block are inherited from the published evaluation by reusing its integral-time estimator verbatim; 240 held-out targets are scored, against 160 in the published run. Intervals are circular moving-block bootstraps at the release and conservative 1.2551τ blocks, the latter governing the registered decision rule, and every contrast reports its across-seed spread. The protocol, arms, estimands, decision rule and cost ceiling were frozen and hash-bound before the production run. The traction is read from the held-out target, so it is an *oracle* traction: the experiment bounds what a closure could transmit through this interface and is not solver-coupled.

4.6 Closure-to-generator composition: fitting and freezing L_A

The direct experiment supplies a traction read from the held-out target; this section closes L_A on the same record. A closure is fitted on the training window only and frozen before any held-out outcome is observed. For every (wall cell, face) pair it reads velocity at wall distances 2.5Δ and 4.5Δ along the face normal—the standard wall-modelled-LES matching-height seam—and never the wall-adjacent cell. Eight dimensionless features are formed: the matching-point y^+; the speed ratio $|\mathbf{u}_t(y_{m2})|/|\mathbf{u}_t(y_{m1})|$; two wall-normal transpiration ratios; the sine and cosine of the skew between the two heights; $\log|\mathbf{u}_t(y_{m1})|$; and a local acceleration proxy $\log(|\mathbf{u}_t(y_{m1})|/\langle|\mathbf{u}_t(y_{m1})|\rangle_{3\times3\times3})$. A six-way face embedding carries geometry, which is known a priori.

With $\mathbf{e}_1 = \mathbf{u}_t(y_{m1})/|\mathbf{u}_t(y_{m1})|$, $\mathbf{e}_2 = \mathbf{n}\times\mathbf{e}_1$, and u_τ^{eq} solving Eq. (7) at y_{m1} by bisection,

$$\widehat{\boldsymbol{\tau}}_w^{\,\mathrm{W}\to\mathrm{F}} = -\,(u_\tau^{\mathrm{eq}})^2\, e^a\left[\cos\theta\,\mathbf{e}_1 + \sin\theta\,\mathbf{e}_2\right], \qquad a = \tfrac{3}{2}\tanh(\cdot), \quad \theta = \tfrac{\pi}{3}\tanh(\cdot), \tag{10}$$

where (a,θ) come from a $14\to64\to64\to2$ multilayer perceptron with zero-initialised output, so training begins *at* the classical equilibrium wall model (evaluated as its own arm, the special case $a=\theta=0$) and departs only where the data pay for it. Because $|\theta|\le\pi/3<\pi/2$, the sign convention $\widehat{\boldsymbol{\tau}}_w\cdot\mathbf{u}_t<0$ holds architecturally for every cell and parameter value. The closure is fitted on snapshots 0–599 against the first-cell traction of Eq. (9), normalised by the training-window traction root-mean-square, for 12,000 AdamW steps and monitored on snapshots 600–659, a validation window *inside* the training span; the terminal state is retained rather than the best-validation checkpoint, so this is validation monitoring, not model selection. It is then frozen by hash and applied unchanged. One stochastic closure fit is shared by all three generator seeds, which therefore probe generator-initialisation robustness conditional on a single closure.

A generator family K is trained from scratch under the same budget as family T with a declared conditioning mixture: oracle first-cell traction 0.40, closure-predicted traction 0.40, zero conditioning 0.20. Both primary traction arms and the absent arm are training classes, so no primary contrast carries a distribution-shift confound. The wrong-information control is a genuine traction from a far-time donor, inside the training distribution, isolating instantaneous correspondence; the mean-load, shuffled and sign-reversed arms are retained as *out-of-training sensitivity probes* carrying no decision-rule clause. Family T's frozen checkpoints are additionally re-evaluated unchanged on the predicted traction. The primary source-excluded region removes every supplied wall cell *and* every cell the closure read, so no cell that carried information into the system is ever scored; its near and outer restrictions use $d/h \le 0.5$ and $d/h > 0.5$, and the four regions of the direct-traction experiment are retained verbatim for identical-definition comparison. The registered decision rule is evaluated on a strict unit: the 103 held-out frames outside the direct-traction experiment's evaluation index—*prospectively unscored, not untouched*: the global contact ledger records that 48 of them carry a score from the earlier equilibrium-adapter experiment. A full unit, the direct-traction experiment's exact 240 frames, is retained for the frozen non-regression table. Intervals are circular moving-block bootstraps at the release and conservative 1.2551τ blocks plus a crossed seed $\times$ time-block interval. The protocol, arms, mixture, controls, metrics, decision rule, tolerances and an eight-GPU-hour ceiling were frozen and hash-bound before the production run; the single preceding software check evaluated training frames only.

4.7 Scores, diagnostics and dependence-aware resampling

For every target and arm, the point prediction is the arithmetic mean of eight generated samples from one realised sampler draw set. With $\boldsymbol{\mu}_{\mathrm{tr}}(\mathbf{r})$ the training-only spatial mean and s_c the training-only scale, standardised fluctuations are $\widetilde{x}_{q,c}(\mathbf{r}) = (x_{q,c}(\mathbf{r}) - \mu_{\mathrm{tr},c}(\mathbf{r}))/s_c$. For region R, the stored sufficient statistics are one sum of squared errors (SSE) and one shared total sum of squares (SST) per physical time,

$$\mathrm{SSE}_{q,a,R} = \sum_{c,\mathbf{r}\in R} (\widetilde{\widehat{x}}_{q,a,c}(\mathbf{r}) - \widetilde{x}_{q,c}(\mathbf{r}))^2, \tag{11}$$

$$\mathrm{SST}_{q,R} = \sum_{c,\mathbf{r}\in R} (\widetilde{x}_{q,c}(\mathbf{r}) - \overline{\widetilde{x}}_R^{\,\mathrm{pool}})^2, \tag{12}$$

and the pooled score is

$$R^2_{\text{fluct}}(a,R) = 1 - \frac{\sum_q \mathrm{SSE}_{q,a,R}}{\sum_q \mathrm{SST}_{q,R}}. \tag{13}$$

For every resampling replicate, Eq. (13) is recomputed for each arm from the same sampled blocks before subtraction; independently bootstrapped limits are never subtracted. Because each region has its own pooled centre and SST, the complete score is not an arithmetic average of the near and farther scores.

The retrieval diagnostic divides every three-component band value by its training-only channel scale, flattens, and for each of the 160 evaluation indices selects $i^\star(q) = \arg\min_{i\le 659} \|\mathbf{x}_{q,B}/\mathbf{s} - \mathbf{x}_{i,B}/\mathbf{s}\|_2^2$, using the associated training field unchanged as the prediction, scored by Eq. (13) with block-selection intervals from 4,000 shared circular draws of 57 samples; it was frozen before execution, with no tuned neighbour count or metric.

For $M = 8$ samples and D standardised degrees of freedom on U_{far}, the multivariate energy score (ES) uses the off-diagonal U-statistic

$$\mathrm{ES} = \frac{1}{M}\sum_{m=1}^{M} \frac{\|\mathbf{X}^{(m)} - \mathbf{y}\|_2}{\sqrt{D}} - \frac{1}{2M(M-1)} \sum_{m\neq \ell} \frac{\|\mathbf{X}^{(m)} - \mathbf{X}^{(\ell)}\|_2}{\sqrt{D}}, \tag{14}$$

lower better [78, 79]; the original evaluation used the $2M^2$ empirical formula, and its retained truth and pairwise distances permit the deterministic correction. The two secondary families are component-spectrum log RMSE (48-point streamwise transforms over the 48 complete $d/h > 0.5$ planes, modes 1–12) and normalised RMSE (NRMSE) of six standardised plane-averaged quadratic-fluctuation profiles ($u'u'$, $v'v'$, $w'w'$, $u'v'$, $u'w'$, $v'w'$)—instantaneous standardised spatial summaries, not conventional dimensional Reynolds stresses. The withdrawn coverage diagnostic compares interpolated 0.1/0.9 quantiles from eight members against 0.8, which is not a finite-ensemble-valid calibration test [80]; block-57 hit fractions are reported descriptively only.

The integral-time convention is $T_{\text{int}}/\Delta t = 1 + 2\sum_{k=1}^{k_0-1} \rho(k)$, with k_0 the first non-positive autocorrelation lag, audited on five signals whose integral times are 39.55, 103.47, 96.30, 122.48 and 77.45 native snapshots; the conservative rule uses the maximum, $T_{\text{int}} = 122.4817$. The 98 omitted fields give $0.800T_{\text{int}}$ and the 1,101-field record has count ratio 8.99. After thinning, a conservative block contains 57 evaluation samples, and M2 sensitivities use 4,000 circular moving-block replicates with shared indices across arms [81, 82]; historical terminal records used a 49-sample block. Moving-block resampling preserves serial structure and pairing without creating independent physical events. Because the evaluation interval was reused for model development, reported 95% intervals describe conditional variation under block selection for a fixed record and configuration; they are not frequentist population-confidence intervals, are not multiplicity-adjusted, and exclude model-selection and between-training-seed uncertainty.

4.8 Periodic-hill protocols

For the early hill support test, all spanwise planes from one time remain together: training, validation and test ranges are 0–479, 500–569 and 590–959, with $N_{\text{eff}} = 3.19$; separate diffusion and flow-matching checkpoints use aligned, absent, random and far-time wall arms. The capacity matrix contains widths 48, 96 and 120 under three fixed seeds at 90,000 updates, resampled by shared circular 116-time blocks crossed with training seeds (10,000 replicates); its preregistered scaling criterion fails.

The two generative families of the generality cells differ in forward process (linear interpolation versus a variance-preserving cosine schedule), training target (velocity versus noise) and sampler (deterministic probability flow versus stochastic ancestral sampling). Their sampler budgets are not matched: each is selected separately by the rule stated below, and the two families converge at different budgets. The grouped retest uses CoNFiLD Case3 (periodic hills, $Re_h = 2800$), rasterised to 64×192 with a curved-lower-wall mask; the corpus is 3 spanwise planes $\times$ 960 physical times stored plane-major, and the experimental unit is the physical time. Train times $[0, 479]$ on all three planes (1,440 arrays); development window $[500, 569]$ on plane 0; decisive unit $[590, 959]$ on plane 0 (370 arrays) with a 110-time gap, plus a robustness subsample of planes 1–2 at every fourth test time; the pipeline asserts pairwise disjointness and aborts otherwise. Viscosity, normalisation, the fluctuation mean and the conditioning scale τ_{rms} come from training times only; the closure's own fit/validation split is by physical time (0–431 fit, 432–479 validate). The only quantity selected from data is the sampler budget, chosen per family on the development window over $\{32, 64, 128, 256\}$ steps using the absent and native-traction arms only (flow matching selects 32; diffusion 256); selecting on the baseline arm cannot favour the closure contrast. Arms are inference-time conditioning swaps over {absent, closure traction, oracle DNS traction, equilibrium-only traction, far-time traction, spatially permuted traction}, with the training mixture (oracle 0.40, closure 0.40, absent 0.20); scoring excludes the conditioning support and the closure's matching-height source cells. Traction is a first-cell viscous shear surrogate on the raster staircase of the lower curved wall, $\boldsymbol{\tau} = -(\nu/d_a)\,\boldsymbol{u}_t$, upper boundary and pressure/form drag excluded. In the band-lift repair, each band cell takes $u_\tau = \sqrt{|\boldsymbol{\tau}|}$ from its nearest wall face and receives $\boldsymbol{u}(d) = u_\tau f_{\text{Reichardt}}(du_\tau/\nu)\,\hat{\boldsymbol{e}}_t$ with zero wall-normal component; band cells coinciding with the closure's matching-height cells are removed from the conditioning support.

One named interval estimator. Every periodic-hill interval in this paper is the same hierarchical bootstrap ($B = 4000$): the three realised seeds are resampled with replacement, then a circular moving-block resample is taken in physical time with block $\lceil 1.2551\tau \rceil$ capped to keep at least three blocks. The measured $\tau = 112.3$ physical times gives block 123 and $N_{\text{eff}} \simeq 3.3$, more conservative than the retained $\tau = 58$; both are reported. Every contrast is recomputed inside each replicate from the same (seed, time) draw before subtraction, including the arm-minus-arm contrasts of Supplementary Note 10, so paired quantities are never formed by subtracting independently bootstrapped limits. An earlier released audit averaged the per-frame sum of squared errors over the realised seeds *before* resampling time; that estimator conditions on the sampler draws and is narrower. Both estimators are reported for every arm, and the declared hierarchical one is quoted throughout. Adopting it changes the zero-exclusion verdict of 7 of 180 arm/region/cell contrasts, always from excluding zero to spanning it, and in five of the seven the affected arm is a control rather than a method arm.

Selection-to-test separation. The grouped contract separates *training* from the decisive unit by 110 physical times. The development window used for sampler-budget and λ selection ends at 569 and the decisive unit begins at 590, a gap of about 20 physical times against an integral time of 112.3. Selection is therefore not separated from the test window in the way training is: it carries strong serial dependence, which a test-set block bootstrap does not cure. No same-time spanwise twin is shared and no selection was revised after test contact, but the two gaps are not interchangeable. The two repairs trained one wall closure and thirteen generators under 8 and 3 GPU-hour ceilings, writing only newly named outputs.

Stochastic sampler for the flow-matching cell. To test whether ensemble under-dispersion *causes* that cell's failure, the deterministic sampler is generalised to the stochastic-interpolant family sharing its marginals [83, 84]. With $\hat{\boldsymbol{z}} = \boldsymbol{x}_t + (1-t)v$ the interpolant's noise estimate—so that $\nabla \log p_t = -\hat{\boldsymbol{z}}/t$—and $\Delta = t - t_n$, the step is $\boldsymbol{x}_{t_n} = \boldsymbol{x}_t - \Delta v - \Delta\lambda\hat{\boldsymbol{z}} + \sqrt{2\lambda t\Delta}\,\boldsymbol{\xi}$, terminating deterministically at $\boldsymbol{x}_t - tv$; at $\lambda = 0$ every added term vanishes and the step is algebraically the released Euler step, so the family is enlarged rather than redefined. No network is retrained: the frozen checkpoints are loaded by hash and only the integration rule changes. λ is selected on the development window over $\{0, 0.5, 2, 8, 32\}$, scoring the absent arm alone, by minimising $|e(\lambda) - 2/(M+1)|$ where e is the rank-histogram extreme-bin mass—a two-sided criterion on calibration, not accuracy, which returns $\lambda = 0$ if no positive value is better calibrated. The test window is contacted once at the selected λ, with a $\lambda = 0$ port-fidelity control. Both candidate stochastic constructions were first checked on an analytically solvable Gaussian problem; the denoising-diffusion implicit model (DDIM) η transport was rejected there for failing to preserve the terminal marginal and for reducing ensemble spread, and was replaced before any outcome on this record was observed (Supplementary Note 10).

4.9 Single-slot channel protocol

The slot experiment uses the smooth-wall channel record of the public CoNFiLD release (nominal $Re_\tau = 180$; measured $Re_\tau = 184$ on the record, the value used throughout), the canonical low-Reynolds-number channel configuration [85]. The generator is trained with exactly one traction-carrying condition: the record's own signed two-component viscous wall traction, corrupted by a variance-preserving fidelity ladder whose mixing coefficient is drawn uniformly per sample, and withheld with probability 0.2. No wall model's output is ever seen in training, so at inference the fitted physics closure (reading two matching-height velocity lines and never any cell below $y^+ \approx 29$), the classical equilibrium law, a trunk- and budget-matched data-only regressor, wrong-instant and spanwise-shuffled tractions, and exact-plus-noise references at four frozen fidelity rungs all enter through the same slot with identical (zero) generator-side training exposure. The conditioning occupies only the four viscous-sublayer rows ($y^+ \leq 3.2$) and is machine-verified absent from every scored cell; at every sampling step those rows are replaced by the sublayer profile implied by the supplied traction, so the generated field's wall gradient equals the supplied traction identically (relative residual

$< 2 \times 10^{-7}$ on every arm). The 46-frame evaluation window had never been scored by any earlier analysis, certified by a machine contact ledger over 127 index sets; every decision rule, control, margin and cost ceiling was frozen in a two-stage preregistration before that window was contacted, and block inference uses three physical-time blocks with a simulation-calibrated critical value $t^* = 4.5$. Both walls and two training seeds are evaluated, and a denoising-diffusion family repeats the primary contrast. The two-phase experiment cost 1.07 GPU-hours; the preregistration, frozen decision applicator and computed outcome are archived with the source data.

4.10 Companion closure, withdrawn diagnostics and quarantine

The frozen companion closure is an empirically trained family-conditioned operator. Its local trunk consumes

$$\mathbf{\Phi} = \left(U_m/U_e,\ y_m/\delta_{99},\ \delta_{99}\partial_x p/U_e^2,\ \delta^*_{\mathrm{res}}\partial_x p/U_e^2\right), \tag{15}$$

while its permutation-invariant branch consumes the member station set $\mathbf{S} = \{U_m/U_e,\ y_m/\delta_{99},\ \delta_{99}\partial_x p/U_e^2,\ \delta^*_{\mathrm{res}}\partial_x p/U_e^2,\ \beta_p\}_{\mathrm{stations}}$, where p is kinematic pressure and the fifth feature is the signed, wall-stress-free pressure-gradient coordinate

$$\beta_p = \frac{\delta^*_{\mathrm{res}}\,\partial_x p}{u_p^2}, \qquad u_p = (\nu|\partial_x p|)^{1/3}, \tag{16}$$

with δ^*_{res} on the resolved interval $y_m \le y \le \delta_{99}$ and $\beta_p = 0$ when $\partial_x p = 0$. The prediction is $\widehat{C}_f = \langle T_\theta(\mathbf{\Phi}), B_\theta(\mathbf{S})\rangle + b$ with $\tau_w = \frac{1}{2}C_f U_e^2$, a signed streamwise kinematic stress under the companion convention, not the wall-on-fluid vector of Eq. (2); "physics-grounded" denotes dimensionless outer inputs with a signed target and stress scaling. Replaying ten retained Krank profiles without fitting quantifies the extractor substitutions: the pressure surrogate has relative RMSE 0.961 and correlation 0.098 against the native pressure gradient, full-wall thickness is 1.292–2.366 times δ^*_{res} (median 1.345), and substitution changes C_f by relative RMSE 0.408, flips one sign, and reduces descriptive all-station R^2 from 0.558 to 0.171; header replay agrees with the export to 2.55×10^{-10}. These facts, not a preference, are why the closure of Sec. 4.6 is fitted on the cube record.

Two withdrawn diagnostics are documented in the Supplementary Information. Case1 has shape $(16001, 918, 3)$ and is a stochastically forced two-dimensional irregular pipe with (u, v, p) [45], not a wall jet; the local pipeline imposed an unsupported obstacle scale, Reynolds number, viscosity and stress and normalised all frames before splitting, so several defects independently preclude attribution (Supplementary Note 8). The archived grouped-time experiment uses a first-cell gradient proxy as its "oracle" and lets the closure consume a subsample of the target field, precluding promotion. Companion geometry-transfer results remain evidence for $\mathcal{C}$ only.

The release entry point verifies sources, rebuilds every released display, compiles both documents and replays available M2 endpoints; a manifest-only clean tree repeats it, and automated checks compare every headline claim and plotted value with source values selected in advance. Legacy 527-frame planar hill results are quarantined because physical-time counterparts crossed the training/test boundary; their derivatives do not support this manuscript. No new flow simulation was performed at any point in this study: every field analysed comes from the frozen LES records. Model training was performed and is declared—seven networks for direct conditioning, one wall closure plus three generators for the closure composition, and the slot experiment's own generators—all written under new names so that no frozen checkpoint is overwritten.

4.11 Computational resources and AI assistance

Model fitting and GPU inference used a single NVIDIA A800 80 GB PCIe GPU. The released environment pins Python 3.10, NumPy 2.2.6, SciPy 1.15.3, Matplotlib 3.10.8, PyTorch 2.10.0, Pillow 12.1.1 and pymech 2.0.1. Experiment-specific GPU-hour costs and ceilings are reported above and in the Supplementary Information; no carbon-footprint estimate was made.

Large language models were used to assist with drafting and editing manuscript text, writing and reviewing analysis and figure-generation code, and organisational consistency checks. Every scientific claim, computation, source-data item, code change and item of final prose was reviewed and approved by the authors, who take full responsibility for the submitted work.

Data availability. Source data for every display and the sufficient statistics that replay the M2 point and ensemble endpoints are deposited in the associated figshare dataset. The exact aligned-cube NekRS case and provenance records are in the repository cited in the abstract. The M0 first-contact result is retained only as a hash-pinned terminal aggregate and cannot be independently reblocked, and the public CoNFiLD Case1 arrays are retained only to audit a withdrawn diagnostic and support no claim. The raw aligned-cube LES volume and the M0/M2 checkpoints are byte-verified locally but too large to distribute with the source-data package; they are available from the corresponding author on request during review and are deposited in a permanent public archive with a DOI on publication. The release licence is fixed at that deposit. The source-data archive also records the provenance of the companion closure-side data; those data do not support the oracle generator endpoint.

Code availability. The review package contains the exact LES case, release-safe reviewed counterparts of the model and evaluation code, deterministic audit derivation, source-data builder, plotting scripts, M2/ensemble and retrieval-sufficient-statistic verifiers, and a manifest-only clean-room rebuild. Exact hashes of the original scripts are documented, and their only differences from the released counterparts are default file paths; the original files require inclusion in the author-approved reviewer archive. The code seeds NumPy and PyTorch, but cuDNN benchmarking is enabled and checkpoints omit random-number-generator state; exact bitwise retraining or continuation is therefore not claimed. The repository cited in the abstract is publicly available at https://github.com/JianouJiang/generative-wall-turbulence. A versioned archival release with a permanent identifier and an explicit open-source licence will accompany publication.

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Acknowledgements. The authors thank the Department of Engineering Science, University of Oxford, for institutional support.

Funding. This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Author contributions. J.J. conceived the study; built the wall closure, the generative models and the evaluation protocols; performed all simulations and analyses; and wrote the manuscript. P.H. critically examined the wall-to-field coupling evidence and validated the figures against their source data. H.Z. critically assessed the scaling evidence. S.J. critically assessed the companion closure-transfer evidence. B.R. supervised the project and guided the wall modelling. All authors reviewed the manuscript.

Competing interests. The authors declare no competing interests.